The Tannakian substrate of GeoVac: a finite-cutoff pro-system, a neutral Tannakian category, and an explicit $\mathbb{G}_a^{3N} \rtimes \mathrm{SL}_2$ inclusion into $\mathrm{Aut}^{\otimes}(\omega)$

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Abstract

We construct the Tannakian substrate of the GeoVac spectral triple at finite cutoff and verify, bit-exact on explicit `sympy`-rational panels, every prerequisite of the Deligne–Milne 1982 neutral Tannakian theorem on an abelian primitive Hopf algebra $\mathcal{H}_{\mathrm{GV}}(n_{\max}) = \mathrm{Sym}_{\mathbb{Q}}(V_{n_{\max}})$ whose primitive space $V_{n_{\max}}$ has $\mathbb{Q}$-dimension $3N(n_{\max})$ (one generator per (sector, Mellin slot) pair). The construction has two halves.

Pro-system half. A cofiltered system of finite-cutoff truncations $\{\mathcal{H}_{\mathrm{GV}}(n_{\max})\}_{n_{\max}\in\mathbb{N}}$ with explicit Hopf-algebra transition maps $\Phi_{m,k} : \mathcal{H}_{\mathrm{GV}}(m) \to \mathcal{H}_{\mathrm{GV}}(k)$ (block-diagonal truncations) satisfies the cofiltered axiom, is U^*-equivariant, admits a well-defined inverse limit $\mathcal{O}_\infty \cong \mathbb{Q}^{\mathbb{N}_{\mathrm{sec}}}$ into which the continuum cochains χ_∞, η_∞ and the continuum Mellin lift $F(s)$ of v3.66.0 FO1–FO3 project coherently, and has finite-cutoff endomorphism algebra $\mathrm{End}_{\mathrm{compat}}(\mathcal{O}_{n_{\max}})$ that is block-lower-triangular of closed-form dimension $\dim = \sum_{j\le i\le n_{\max}}(i+1)(j+1)$.

Tannakian half. The finite-dimensional representation category $\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max}))$ is an abelian, symmetric monoidal, rigid $\mathbb{Q}$-linear category (axioms TC-1a, TC-1b, TC-1c bit-exact); the no-op forgetful $\omega :\ \mathrm{Rep}_{\mathrm{fin}} \to \mathbf{Vec}_{\mathbb{Q}}$ is a faithful exact $\otimes$-functor (TC-1d); and the pro-unipotent factor of the cosmic Galois group $U^*_{\mathrm{Levi}} = \mathbb{G}_a^{3N(n_{\max})} \rtimes \mathrm{SL}_2$ admits an *explicit inclusion*

$$\Phi :\ U^*_{\mathrm{Levi}}(\mathbb{Q}) \longrightarrow \mathrm{Aut}^{\otimes}(\omega), \qquad \Phi(t,g)(V) = \exp\Big(\sum_{g\in\mathfrak{P}} t_g X_g^V\Big) \otimes \rho_{V_{\mathrm{PW}}}(g)$$

on the natural-substrate panel $V \otimes V_{\mathrm{PW}}$, which is bit-exact at every point of the verification grid (TC-1e + TC-1f). At $n_{\max} = 2$ the inclusion is *full-panel injective*: for every one of the $3 \cdot N(2) = 15$ primitive generators g a 2-dim test rep V_g detects g uniquely with $\eta_{V_g}(t) = I$ iff $t_g = 0$.

Total bit-exact verification panel: 5,864 *zero residuals* (`sympy`-rational throughout) across PS-1/2/3/4 + TC-1a/b/c/d/e/f + TC-2a/b/c/d + G4-Inj at $n_{\max} \in \{2, 3, 4\}$. The G4-Inj C4 sub-block is panel-size bookkeeping for the *abelianized* image: its *full* M3-column injectivity is *refuted* (the χ_{-4}-enriched Gram is rank $2 \ll N$), so those residuals do not certify a closed immersion (Theorem 6.2). The fourteenth math.OA standalone in the GeoVac series.

Cosmic-Galois injection. We close the *injection direction* $U^*_{GV} \to \mathcal{U}_4 \rtimes SL_2$ into the (abelianization of the) unipotent radical of the level-4 motivic Galois group $\mathcal{G}_4 = \mathcal{G}_{\mathrm{MT}(\mathbb{Z}[i,1/2])}$ of Deligne 2010 + Glanois 2015, with four structural compatibilities (multiplicativity, depth-1 coproduct, SL_2-to-Levi via cyclotomic χ_4, abelianized image via Glanois basis; Section 6,

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Theorem 6.2). The proof uses the Eskandari–Murty–Nemoto 2025 level-4 placement of G + the Goncharov–Deligne 2005 faithful action on $\pi_1^{\mathrm{mot}}(\mathbb{G}_m - \mu_4)$. Equality / surjectivity on the depth-≥ 2 content is the named long-range forward research question.

Honest scope. We close the *inclusion* direction $\mathrm{Aut}^{\otimes}(\omega) \supseteq U^*_{\mathrm{Levi}}$ at finite cutoff with both Levi factors explicit (TC-1e/TC-1f). The *converse equality* $\mathrm{Aut}^{\otimes}(\omega) = U^*_{\mathrm{Levi}}$ at finite cutoff is closed by the TC-2 sub-arc: TC-2a (abelian factor at $n_{\max} = 2$), TC-2b (SL_2 factor on the Peter–Weyl panel), TC-2c (higher cutoffs $n_{\max} \in \{3, 4\}$), and TC-2d (pro-system cofiltered coherence on the natural-substrate panel). The remaining long-range content is the *inverse-limit equality* of Φ^{inj} — the pro-finite Tannakian theorem coherent with v3.66.0 FO3 Interpretation C on $\mathcal{O}_\infty$ (PS-3) and with exhaustion of every level-4 cyclotomic motivic MZV in the GeoVac master Mellin engine output.

Contents

1 Introduction

1.1 The question

The GeoVac spectral triple is constructed on a finite combinatorial graph at every cutoff $n_{\max} \in \mathbb{N}$: a Coulomb graph on Fock-projected S^3 (Paper 7) carrying a Camporesi–Higuchi spinor bundle with half-integer-shifted Dirac spectrum (Paper 32). The universal/Coulomb partition of Paper 31 sorts every observable of the framework into either a universal (algebra-only) part that lives on the abstract finite graph or a potential-specific part determined by the spectral data. After the Mixed-Tate Test of v3.46.0 + the Periods-A arc of v3.48.0 + the Q5'-HardParts arc of v3.61.0–v3.66.0 the structural reading of the universal sector crystallised: each of the three master-Mellin engine sub-mechanisms M1 / M2 / M3 lives in a cyclotomic mixed-Tate period ring at level ≤ 4 over $\mathbb{Z}[i, 1/2]$; the natural categorical home for these periods is a (motivic) Galois group, and the natural candidate is a pro-unipotent affine group scheme U^* with Levi decomposition $U^*_{\text{Levi}} = \mathbb{G}_a^{3N} \rtimes \mathrm{SL}_2$ at finite cutoff (v3.63.0 Track 1).

The present paper asks the next structural question: is the substrate for a *neutral Tannakian theorem* on the underlying Hopf-algebra category fully assembled at finite cutoff? Concretely:

- Is there a cofiltered pro-system $\{\mathcal{H}_{\text{GV}}(n_{\max}), \Phi_{m,k}\}$ of Hopf algebras whose inverse limit carries the period content?
- Is the representation category $\mathrm{Rep}_{\text{fin}}(\mathcal{H}_{\text{GV}}(n_{\max}))$ abelian, symmetric monoidal, and rigid (the three Deligne–Milne 1982 prerequisites for being neutral Tannakian)?
- Is there a fiber functor $\omega : \ \mathrm{Rep}_{\text{fin}} \to \mathbf{Vec}_{\mathbb{Q}}$?
- Does the candidate $U^*_{\text{Levi}} = \mathbb{G}_a^{3N} \rtimes \mathrm{SL}_2$ admit an *explicit inclusion* into $\mathrm{Aut}^{\otimes}(\omega)$?

This paper answers *yes* to all four, bit-exact at finite cutoff, on the abelian primitive substrate $\mathcal{H}_{\text{GV}}(n_{\max}) = \mathrm{Sym}_{\mathbb{Q}}(V_{n_{\max}})$ where $V_{n_{\max}}$ has $\mathbb{Q}$-dimension $3N(n_{\max})$.

1.2 What this paper does not claim

The Tannakian reconstruction direction $\mathrm{Aut}^{\otimes}(\omega) = U^*_{\text{Levi}}$ at finite cutoff is closed in Sections 5.5, 5.6, 5.7, and 5.8 as the TC-2 sub-arc (Theorems 5.7, 5.9, 5.11, 5.13). What is *not* closed by this paper is the *inverse-limit identification* of $\mathrm{Aut}^{\otimes}(\omega)^{(\infty)} = \mathbb{G}_a^{\infty} \rtimes SL_2$ with the motivic Galois group on the mixed-Tate period ring of v3.66.0 FO3 Interpretation C on $\mathcal{O}_\infty$ (PS-3) — this requires the pro-finite Tannakian theorem [2, 3] coherent with the period content and remains the named long-range frontier.

We also do *not* introduce any new ring of periods, any new spectral object, or any new conjecture about α. The Hopf algebra $\mathcal{H}_{\text{GV}}$ is the v3.61.0 Track A Hopf candidate; the period content is the v3.66.0 FO1–FO3 + Paper 55 classification; the cosmic-Galois candidate U^*_{Levi} is the v3.63.0 Track 1 Levi decomposition. This paper consolidates the substrate for these existing pieces into one bit-exact assembly.

1.3 Roadmap

Section 2 constructs the pro-system half (PS-1 transitions + PS-2 U^*-equivariance + PS-3 inverse limit + PS-4 endomorphism rigidity). Section 3 verifies that $\mathrm{Rep}_{\mathrm{fin}}$ is abelian, symmetric monoidal, and rigid (TC-1a/b/c). Section 4 constructs the fiber functor (TC-1d). Section 5 establishes the explicit $\mathbb{G}_a^{3N} \rtimes \mathrm{SL}_2$ inclusion into $\mathrm{Aut}^{\otimes}(\omega)$ (TC-1e + TC-1f), plus the TC-2 converse-direction sub-arc closing the finite-cutoff reconstruction in Sections 5.5 (abelian factor), 5.6 (SL_2 factor), 5.7 (higher cutoffs), and 5.8 (cofiltered coherence). Section 6 closes the cosmic-Galois injection direction $U^*_{GV} \to \mathcal{U}_4 \rtimes SL_2$ into the (abelianization of the) unipotent radical of the level-4 motivic Galois group $\mathcal{G}_4$ (Theorem 6.2). Section 7 catalogues the 5,864-residual bit-exact verification panel. Section 8 states the open long-range frontier (reconstruction) and the natural sprint-scale follow-ons (higher $n_{\max}$ injectivity panels, SL_2 on higher Peter–Weyl reps, non-natural-substrate enlargements).

1.4 Related work

GeoVac's gauge-structure scaffolding is the Marcolli–van Suijlekom 2014 gauge-network construction [10] as sharpened by Perez-Sanchez 2024/2025 [11, 12] — the Standard Model recovered from the spectral action without a Higgs (matching the GeoVac Sprint H1 Yukawa-non-selection theorem). *Honest scope of the MvS-lineage claim:* GeoVac is an *instance* of MvS 2014 (with the Perez-Sanchez 2025 correction applied) at the data level (graph, vertex spectral triple, edge connection, spectral action). It is an *extension* of MvS at the structural level: the forced-graph claim (Bertrand × Fock projection), the master Mellin engine, the explicit Hopf-algebra substrate $\mathcal{H}_{\mathrm{GV}}$, and the Tannakian dual U^* constructed here are GeoVac-original. MvS 2014 does *not* predict a Tannakian dual; the cosmic-Galois layer U^*_{Levi} sits on top of the gauge-network scaffolding as additional structure, not as an MvS corollary.

The neutral-Tannakian backbone we transport is Deligne–Milne 1982 [1], the modern motivic-Galois treatment is André 2004 [4] and Brown [5], the cyclotomic mixed-Tate package is Deligne 2010 [7] + Glanois 2015 [8], and the Connes–Marcolli cosmic Galois group U^*_{CM} is [9]. We flag one common framing slip: the two groups are related by a *surjection* $U^*_{CM} \twoheadrightarrow \mathcal{G}_{MT(\mathbb{Z})}$, not an identification. This is an elementary comparison of two free pro-unipotent groups, not a theorem we are importing: U^*_{CM} has one generator in every positive degree (free non-abelian, Connes–Marcolli [9]) while $\mathcal{G}_{MT(\mathbb{Z})}$ has generators only in odd degrees ≥ 3 (Brown 2012 [5]). Brown does *not* assert the comparison; his cosmic-Galois paper explicitly leaves open whether his groups are related to the Connes–Marcolli ones at all. The period ring of GeoVac sits inside $\mathrm{MT}(\mathbb{Z}[i, 1/2])$ at level ≤ 4; see Paper 55 [23] for the cyclotomic-mixed-Tate classification. The natural cosmic-Galois comparison target for U^*_{GV} is therefore $\mathcal{G}_{MT(\mathbb{Z}[i,1/2])} = \mathcal{G}_4$, the level-4 motivic Galois group (Deligne 2010), *not* Brown's smaller $\mathcal{G}_{MT(\mathbb{Z})}$ (Eskandari–Murty–Nemoto 2025 [25] place Catalan G at level 4; that G is *not* a period of $\mathrm{MT}(\mathbb{Z})$ is expected but not proven, so the refinement is motivated by where G is known to live, not compelled by a proven exclusion). Bost–Connes / Greenfield–Marcolli–Teh QSM lineage is *ruled out* as a transport target on four structural grounds (see Paper 55 §subsec:open_m2_m3 honest-scope paragraph and the Greenfield–Marcolli–Teh transport memo); the substrate built here is neutral Tannakian on a finite-dimensional symmetric algebra, not QSM.

2 The pro-system half

2.1 The cutoff Hopf algebra $\mathcal{H}_{\mathrm{GV}}(n_{\max})$

For each integer $n_{\max} \geq 1$ let $\mathfrak{S}(n_{\max}) = \{(n, l) : 1 \leq n \leq n_{\max}, 0 \leq l \leq n\}$ be the set of *Coulomb sectors at cutoff* $n_{\max}$ (the construction's convention, matching `geovac/pro_system.py`), $N_{\mathrm{sec}}(n_{\max}) = |\mathfrak{S}(n_{\max})| = n_{\max}(n_{\max}+3)/2$. The *primitive space* of $\mathcal{H}_{\mathrm{GV}}(n_{\max})$ is the $\mathbb{Q}$-vector space

$$V_{n_{\max}} = \bigoplus_{(n,l)\in\mathfrak{S}(n_{\max})} \bigoplus_{k\in\{0,1,2\}} \mathbb{Q}\cdot x_{(n,l),k},$$

of $\mathbb{Q}$-dimension $3N_{\mathrm{sec}}(n_{\max})$, where $k \in \{0,1,2\}$ is the *master Mellin slot*: $k = 0$ for the Hopf-base measure mechanism M1, $k = 1$ for vertex-parity Hurwitz M3, $k = 2$ for Seeley–DeWitt M2 (Paper 18 §III.7; Paper 32 §VIII).

Definition 2.1. The *GeoVac Hopf algebra at cutoff* $n_{\max}$ is $\mathcal{H}_{\mathrm{GV}}(n_{\max}) = \mathrm{Sym}_{\mathbb{Q}}(V_{n_{\max}})$, the symmetric algebra of the primitive space, with coproduct $\Delta x = x\otimes 1 + 1\otimes x$ on generators (so every generator is primitive), counit $\varepsilon x = 0$, antipode $Sx = -x$, and extended multiplicatively. This is the v3.61.0 Track A candidate Hopf algebra.

$\mathcal{H}_{\mathrm{GV}}(n_{\max})$ is a primitively generated commutative Hopf algebra over $\mathbb{Q}$ (a polynomial Hopf algebra). Its group of group-like elements is trivial; its Lie algebra of primitive elements is the abelian Lie algebra $V_{n_{\max}}$.

2.2 Transition maps and the cofiltered axiom (PS-1)

Definition 2.2. For $m \geq k \geq 1$ let $\Phi_{m,k} : \mathcal{H}_{\mathrm{GV}}(m) \to \mathcal{H}_{\mathrm{GV}}(k)$ be the block-diagonal Hopf-algebra truncation that sends a generator $x^{(m)}_{(n,l),s}$ to $x^{(k)}_{(n,l),s}$ if $n \leq k$ and to 0 otherwise, extended multiplicatively.

Theorem 2.3 (Cofiltered axiom, PS-1)**.** *For every $m \geq k \geq j \geq 1$ the composition $\Phi_{k,j}\circ\Phi_{m,k}$: $\mathcal{H}_{\mathrm{GV}}(m) \to \mathcal{H}_{\mathrm{GV}}(j)$ equals the direct transition $\Phi_{m,j} : \mathcal{H}_{\mathrm{GV}}(m) \to \mathcal{H}_{\mathrm{GV}}(j)$ bit-exact on every element of $\mathcal{H}_{\mathrm{GV}}(m)$.*

Proof. On a primitive generator $x^{(m)}_{(n,l),s}$, both sides act as truncation by $n \leq j$: $\Phi_{m,j}(x) = x^{(j)}_{(n,l),s}$ if $n \leq j$, else 0; $\Phi_{k,j}(\Phi_{m,k}(x))$ first truncates by $n \leq k$ then by $n \leq j$, which collapses to $n \leq j$ since $j \leq k$. Extension to $\mathcal{H}_{\mathrm{GV}}(m) = \mathrm{Sym}_{\mathbb{Q}}(V_m)$ is via the multiplicative structure, preserved by definition. Bit-exact verification at all 10 triples in $\{1,\ldots,5\}$ confirms the identity at the matrix level. See PS-1 sprint memo. □ □

2.3 U^*-equivariance (PS-2)

The cosmic-Galois candidate U^*_{Levi} acts on $\mathcal{H}_{\mathrm{GV}}(n_{\max})$ at the primitive level via the Levi decomposition $\mathbb{G}_a^{3N(n_{\max})} \rtimes \mathrm{SL}_2$. The pro-unipotent $\mathbb{G}_a^{3N(n_{\max})}$ factor acts as $x \mapsto x + t_x\cdot 1$ on each generator (translation, where $t_x \in \mathbb{Q}$); the SL_2 factor acts on the Mellin slot via a $\mathbb{Q}$-linear action on the $k = 0,1,2$ index (the v3.63.0 substrate explicitly identifies the SL_2 factor as acting on the $j_{\max}$ axis of the truncation, with Peter–Weyl decoration on the representations realising the action; see §5.2).

Theorem 2.4 (U^*-equivariance, PS-2)**.** *The transition $\Phi_{m,k}$ is U^*-equivariant: for every $t = (t_x) \in \mathbb{Q}^{3N(m)}$ and every $g \in \mathrm{SL}_2(\mathbb{Q})$, the identity*

$$\Phi_{m,k}\circ\Phi(t,g) \;=\; \Phi(t_{|k},g)\circ\Phi_{m,k} \quad \text{on } \mathcal{H}_{\mathrm{GV}}(m),$$

holds bit-exact. Here $t_{|k}$ is the restriction of t to the primitive generators of $\mathcal{H}_{\mathrm{GV}}(k)$. Verified at 435 $\mathbb{G}_a$ generator compatibility cells +40 class-level χ,η compatibility cells; SL_2 *commutativity is categorical (independence of axes).*

2.4 The inverse limit $\mathcal{O}_\infty$ (PS-3)

Let $\mathcal{O}_{n_{\max}}$ be the (abelian) class algebra of $\mathcal{H}_{\mathrm{GV}}(n_{\max})$ at cutoff $n_{\max}$ (one class per sector). The transition $\Phi_{m,k}$ restricts to a sector-truncation $\mathcal{O}_m \to \mathcal{O}_k$. Set

$$\mathcal{O}_\infty = \varprojlim_{n_{\max}} \mathcal{O}_{n_{\max}}.$$

Theorem 2.5 (Inverse limit, PS-3). *$\mathcal{O}_\infty \cong \mathbb{Q}^{\mathbb{N}_{\mathrm{sec}}}$, the $\mathbb{Q}$-vector space of finitely-supported sequences over the natural-number-indexed sector set $\mathbb{N}_{\mathrm{sec}} = \{(n,l) : \ n \geq 1, 0 \leq l \leq n\}$ (shell n therefore has dimension $n+1$, giving $N(n_{\max}) = n_{\max}(n_{\max}+3)/2$; this is the convention `geovac/pro_system.py` enumerates, and the one Theorem 2.6's $(i+1)\times(j+1)$ blocks require). The continuum cochains χ_∞, η_∞ of v3.59.0 (Mellin M2-residue $\sqrt{\pi}/2$ + sector-LOCAL closed forms) project coherently to $\mathcal{O}_\infty$: for every $n_{\max}$, the projection $\pi_{n_{\max}} : \mathcal{O}_\infty \to \mathcal{O}_{n_{\max}}$ sends χ_∞ to $\chi_{n_{\max}}$, and similarly for η_∞. The continuum Mellin lift $F(s)$ of v3.65.0 T1 (cf. v3.66.0 FO2) projects with weight/depth grading on MT($\mathbb{Q}$, 1) onto each $\mathcal{O}_{n_{\max}}$.*

The proof is the universal property of the inverse limit applied to the cofiltered system $\{\Phi_{m,k}\}$ of Theorem 2.3, together with the bit-exact projection identities of PS-3 (284 residuals on the universal property and continuity-under-transitions panels).

2.5 Endomorphism rigidity (PS-4)

Theorem 2.6 (Endomorphism rigidity, PS-4). *Let $\mathrm{End}_{\mathrm{compat}}(\mathcal{O}_{n_{\max}})$ denote the algebra of $\mathbb{Q}$-linear endomorphisms of $\mathcal{O}_{n_{\max}}$ that intertwine all transitions $\Phi_{m,k}$ for $m,k \leq n_{\max}$. Then $\mathrm{End}_{\mathrm{compat}}(\mathcal{O}_{n_{\max}})$ is the block-lower-triangular subalgebra of $\mathrm{End}_{\mathbb{Q}}(\mathcal{O}_{n_{\max}})$ with respect to the natural sector grading, of closed-form dimension*

$$\dim_{\mathbb{Q}} \mathrm{End}_{\mathrm{compat}}(\mathcal{O}_{n_{\max}}) \ = \sum_{1\leq j\leq i\leq n_{\max}} (i+1)(j+1) \ = \ \frac{n_{\max}(n_{\max}+1)(n_{\max}+2)(3n_{\max}+13)}{24} \quad (= 19, 55, 125, 2$$

The closed form is a symbolic identity—proved, and regression-pinned, in `tests/test_paper56_endo_rigidity`

Remark 2.7. The Tannakian-naive expectation would be that $\mathrm{End}_{\mathrm{compat}}(\mathcal{O}_{n_{\max}}) = \mathbb{Q}\cdot\mathrm{id}$ (rigid scalar diagonal). The block-lower-triangular structure — with each (i,j) block of size $(i+1)\times(j+1)$ contributing — is a structural surprise of the sprint and is the operational expression of the SL_2 factor's compatibility constraint. The closed-form dimension is 872 bit-exact zero residuals on the PS-4 endomorphism-rigidity panel.

2.6 Pro-system half summary

Sub-track	Statement	Residuals
PS-1	Transitions $\Phi_{m,k}$ + cofiltered axiom	130
PS-2	U^*-equivariance ($\mathbb{G}_a$ + class-level)	485
PS-3	Inverse limit + continuity	284
PS-4	Endomorphism rigidity	872
Total pro-system half		**1,771**

Table 1: Pro-system half: 1,771 bit-exact zero residuals across the four lockdown sub-tracks.

3 The representation category $\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max}))$

3.1 Finite-dimensional Hopf representations

Definition 3.1. A *finite-dimensional representation* V of $\mathcal{H}_{\mathrm{GV}}(n_{\max})$ over $\mathbb{Q}$ is a pair $(\underline{V}, \{X_g^V\})$ where $\underline{V}$ is a finite-dim $\mathbb{Q}$-vector space and $\{X_g^V \in \mathrm{End}_{\mathbb{Q}}(\underline{V})\}_{g\in\mathfrak{P}}$ is a family of *pairwise commuting nilpotent* endomorphisms indexed by the primitive generator set $\mathfrak{P}$ of $\mathcal{H}_{\mathrm{GV}}(n_{\max})$. The action of $\mathcal{H}_{\mathrm{GV}}(n_{\max})$ on $\underline{V}$ is by the algebra homomorphism $x_g \mapsto X_g^V$, extended multiplicatively (well-defined because the X_g^V commute).

The category $\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max}))$ has these representations as objects and *intertwiners* $f : V \to W$ (i.e. $\mathbb{Q}$-linear maps satisfying $f \circ X_g^V = X_g^W \circ f$ for every g) as morphisms.

3.2 Abelian (TC-1a)

Theorem 3.2 (TC-1a). *$\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max}))$ is an abelian $\mathbb{Q}$-linear category in the Deligne–Milne 1982 sense: it has a zero object, finite direct sums, kernels and cokernels, every monomorphism is the kernel of its cokernel, and every epimorphism is the cokernel of its kernel.*

The proof inherits the abelian structure of $\mathbf{Vec}_{\mathbb{Q}}$ at the level of underlying vector spaces and checks that the kernel / cokernel of an intertwiner inherits a commuting-nilpotent endomorphism family — which it does, because the X_g^V are sub- and quotient-stable under intertwiners. Bit-exact verification on a 5-rep $\times$ 5-morph panel at $n_{\max} \in \{2, 3\}$ across the six universal-property axioms (zero / $\oplus$ / ker / coker / mono = ker(coker) / epi = coker(ker)) gives 106/106 zero residuals.

3.3 Symmetric monoidal (TC-1b)

Theorem 3.3 (TC-1b). *$\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max}))$ is a strict symmetric monoidal category with unit object the trivial 1-dim rep $\mathbf{1}$ (all $X_g^{\mathbf{1}} = 0$).*

The tensor structure is diagonal: $V \otimes W$ has underlying space $\underline{V} \otimes_{\mathbb{Q}} \underline{W}$ and endomorphisms $X_g^{V\otimes W} = X_g^V \otimes I_{\underline{W}} + I_{\underline{V}} \otimes X_g^W$ (the cocommutative Hopf-algebra coproduct). Associativity is the identity in lex order; braiding is the swap permutation $V \otimes W \to W \otimes V$, symmetric. Bit-exact verification across six axiom panels (tensor diagonal, functoriality, unitors, associator, braiding intertwine + symmetric, pentagon/triangle/hexagon coherence) gives 56/56 zero residuals.

3.4 Rigid (TC-1c)

Theorem 3.4 (TC-1c). *$\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max}))$ is rigid: every V has a dual $V^\vee$ with evaluation $\mathrm{ev} : V^\vee \otimes V \to \mathbf{1}$ and coevaluation $\mathrm{coev} : \mathbf{1} \to V \otimes V^\vee$ satisfying the snake identities.*

The dual is constructed via the contragredient action: $V^\vee$ has underlying space the $\mathbb{Q}$-dual $\underline{V}^*$, with endomorphisms $X_g^{V^\vee} = -(X_g^V)^T$. The evaluation and coevaluation are the standard pairings on $\mathbf{Vec}_{\mathbb{Q}}$; they intertwine the X_g actions because $X_g^{V^\vee \otimes V} + X_g^{V\otimes V^\vee}$ vanishes on the image of coev and on the kernel of ev.

Remark 3.5 (Strict double dual). On the abelian primitive substrate $(V^\vee)^\vee = V$ is a *strict equality*, not merely a canonical isomorphism: $X_g^{(V^\vee)^\vee} = -(-X_g^{VT})^T = X_g^V$, and the double-dual underlying space identification $\underline{V}^{**} = \underline{V}$ holds strictly on the finite-dim panel. The Hopf antipode squares to identity, $S^2 = \mathrm{id}$, in the symmetric algebra of an abelian Lie algebra, which is the structural reason. **Scope:** this strict equality should not be claimed beyond the abelian primitive substrate $\mathcal{H}_{\mathrm{GV}} = \mathrm{Sym}_{\mathbb{Q}}(V)$.

Bit-exact verification gives 50/50 zero residuals on the rigidity panel.

4 The fiber functor (TC-1d)

Definition 4.1. The *fiber functor* $\omega : \mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max})) \to \mathbf{Vec}_{\mathbb{Q}}$ sends $V = (\underline{V}, \{X_g^V\})$ to $\underline{V}$ (the underlying $\mathbb{Q}$-vector space) and an intertwiner f to f itself.

Theorem 4.2 (TC-1d)**.** *ω is a faithful exact $\mathbb{Q}$-linear $\otimes$-functor* $\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max})) \to \mathbf{Vec}_{\mathbb{Q}}$*:*

1. *(exact)* $\omega(\ker f) = \ker \omega(f)$ *and* $\omega(\operatorname{coker} f) = \operatorname{coker} \omega(f)$ *bit-exact (the kernel/cokernel calculation in* $\mathrm{Rep}_{\mathrm{fin}}$ *is the kernel/cokernel in* $\mathbf{Vec}_{\mathbb{Q}}$ *on underlying spaces);*
2. *(faithful)* $f \neq 0$ *in* $\mathrm{Rep}_{\mathrm{fin}}$ *iff* $\omega(f) \neq 0$ *in* $\mathbf{Vec}_{\mathbb{Q}}$ *(intertwiners are determined by their underlying linear maps);*
3. *($\otimes$-preserving)* $\omega(V \otimes W) = \omega(V) \otimes_{\mathbb{Q}} \omega(W)$ *strict;*
4. *(unit-preserving)* $\omega(\mathbf{1}) = \mathbb{Q}$.

Bit-exact verification across the four properties + a faithfulness witness panel gives 40/40 zero residuals.

Remark 4.3. The fiber functor here is the *no-op forgetful*: a finite-dimensional Hopf representation IS its underlying vector space plus the Hopf-data decoration $\{X_g^V\}$, and ω forgets the decoration. This is the natural choice for an abelian primitive Hopf algebra over $\mathbb{Q}$; in the Deligne–Milne 1982 vocabulary ω is a $\mathbb{Q}$-rational fiber functor and the resulting Tannakian category is *neutral*.

5 The explicit $\mathbb{G}_a^{3N} \rtimes \mathrm{SL}_2$ inclusion

5.1 The pro-unipotent factor (TC-1e)

Theorem 5.1 (TC-1e)**.** *The map*

$$\Phi : \mathbb{Q}^{3N(n_{\max})} \to \mathrm{Aut}^{\otimes}(\omega), \qquad \Phi(t)(V) = \exp\Bigl(\sum_{g \in \mathfrak{P}} t_g X_g^V\Bigr) = \eta_V(t),$$

is a well-defined natural $\otimes$-automorphism of ω for every $t \in \mathbb{Q}^{3N(n_{\max})}$, and a group homomorphism into $\mathrm{Aut}^{\otimes}(\omega)$.

Proof sketch. The matrix exponential terminates exactly because $\sum_g t_g X_g^V$ is a $\mathbb{Q}$-linear combination of pairwise commuting nilpotent endomorphisms, hence nilpotent. The four Deligne–Milne 1982 natural $\otimes$-automorphism axioms reduce to:

- *Invertibility.* $\eta_V(t)^{-1} = \eta_V(-t)$, since exp is a group homomorphism from $\mathbb{Q}^{3N(n_{\max})}$ to $\mathrm{GL}(\underline{V})$ on a commuting nilpotent input.
- *Unit.* $\eta_{\mathbf{1}}(t) = \exp(0) = I$ on the trivial rep (all $X_g^{\mathbf{1}} = 0$).
- *Naturality.* For an intertwiner $f : V \to W$, $f \circ X_g^V = X_g^W \circ f$ extends to $f \circ \eta_V(t) = \eta_W(t) \circ f$ by termwise application of the exp power series.
- *$\otimes$-compatibility.* $\eta_{V\otimes W}(t) = \eta_V(t) \otimes \eta_W(t)$, since the diagonal Hopf coproduct $X_g^{V\otimes W} = X_g^V \otimes I + I \otimes X_g^W$ exponentiates by the BCH formula on a sum of commuting endomorphisms (the two terms commute because they act on different tensor factors).

The group-law condition $\Phi(t_1 + t_2) = \Phi(t_1) \cdot \Phi(t_2)$ is the commuting-nilpotent BCH again. Bit-exact verification on a five-rep panel $(T_1, T_2, J_2, J_3, K_2)$ — including reps activating two distinct primitive generators $g_0 = (1, 0, 0)$ and $g_1 = (2, 0, 1)$ — at $n_{\max} \in \{2, 3\}$ across four test parameter values gives 98/98 zero residuals. □ □

5.2 The SL_2 factor (TC-1f)

Theorem 5.2 (TC-1f, SL_2 axioms on Peter–Weyl panel). *On the Peter–Weyl panel*

$$\mathfrak{PW} = \{V_{\mathrm{triv}},\ V_{\mathrm{fund}} = \mathbb{Q}^2,\ \mathrm{Sym}^2 V_{\mathrm{fund}} = \mathbb{Q}^3\}$$

with the standard $\mathrm{SL}_2(\mathbb{Q})$ *action* $\rho :\ \mathrm{SL}_2(\mathbb{Q}) \to \mathrm{GL}(V_{\mathrm{PW}})$ *(identity on* V_{triv}*;* $g \mapsto g$ *on* V_{fund}*;* $g \mapsto \mathrm{Sym}^2(g)$ *on* $\mathrm{Sym}^2 V_{\mathrm{fund}}$*), the three structural identities* ρ *is required to satisfy — invertibility,* $\otimes$*-compatibility, group homomorphism — are bit-exact for every* g *in the five-element panel* $\{e, u, t, s, w\} = \{$*identity, unipotent* $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$*, torus* $\mathrm{diag}(2, 1/2)$*, generic* $\left(\begin{smallmatrix} 5 & 2 \\ 7 & 3 \end{smallmatrix}\right)$*, Weyl involution* $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)\}$.

Theorem 5.3 (TC-1f, $\mathbb{G}_a \times \mathrm{SL}_2$ commute on combined reps). *For every* $t \in \mathbb{Q}^{3N(n_{\max})}$*, every* $g \in \mathrm{SL}_2(\mathbb{Q})$*, every* $V \in \mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max}))$*, and every* $V_{\mathrm{PW}} \in \mathfrak{PW}$*, the two automorphisms of* $\omega(V \otimes V_{\mathrm{PW}}) = \underline{V} \otimes_{\mathbb{Q}} \underline{V_{\mathrm{PW}}}$ *commute:*

$$(\eta_V(t) \otimes I)(I \otimes \rho_{V_{\mathrm{PW}}}(g)) \;=\; \eta_V(t) \otimes \rho_{V_{\mathrm{PW}}}(g) \;=\; (I \otimes \rho_{V_{\mathrm{PW}}}(g))(\eta_V(t) \otimes I).$$

Proof. Both sides are the Kronecker product $\eta_V(t) \otimes \rho_{V_{\mathrm{PW}}}(g)$ by the Kronecker identity $(A \otimes I)(I \otimes B) = A \otimes B = (I \otimes B)(A \otimes I)$, which is the structural witness that $\mathbb{G}_a^{3N}$ and SL_2 act on disjoint tensor factors of the combined rep. The combined automorphism is $\Phi(t, g)(V \otimes V_{\mathrm{PW}}) = \eta_V(t) \otimes \rho_{V_{\mathrm{PW}}}(g)$. □ □

Bit-exact verification on 100 (parameter, g, V, V_{PW}) cells gives 100/100 zero residuals.

5.3 Full-panel injectivity at $n_{\max} = 2$ (TC-1f)

Theorem 5.4 (TC-1f, full-panel injectivity). *At* $n_{\max} = 2$*, for every one of the* $3 \cdot N(2) = 15$ *primitive generators* $g \in \mathfrak{P}_{n_{\max}=2}$ *there is a 2-dim test representation* $V_g \in \mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(2))$ *with*

$$X_g^{V_g} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad X_h^{V_g} = 0 \text{ for every } h \neq g,$$

such that

$$\eta_{V_g}(t) = I \quad \Longleftrightarrow \quad t_g = 0.$$

Consequently the inclusion $\Phi :\ \mathbb{Q}^{3N(2)} \to \mathrm{Aut}^{\otimes}(\omega)$ *is injective at* $n_{\max} = 2$*: if* $t \neq 0$ *then* t *has some* $t_{g_*} \neq 0$ *and* $\eta_{V_{g_*}}(t) \neq I$*, so* $\Phi(t) \neq \mathrm{id}_\omega$.

Proof. $\eta_{V_g}(t) = \exp(t_g X_g^{V_g}) = I + t_g X_g^{V_g}$ since $(X_g^{V_g})^2 = 0$. The $(1, 2)$ entry of $\eta_{V_g}(t) - I$ equals t_g, which is zero iff $t_g = 0$. Bit-exact verification on the full 255-cell panel (15 zero-t cells + 15 single-t activations × 15 test reps + 15 generic-t cells) gives 255/255 zero residuals. □ □

5.4 Inclusion summary

Corollary 5.5 (TC-1e + TC-1f). *At every* $n_{\max} \geq 1$ *there is an explicit group homomorphism*

$$\Phi :\ U^*_{\mathrm{Levi}}(\mathbb{Q}) = \mathbb{G}_a^{3N(n_{\max})}(\mathbb{Q}) \rtimes \mathrm{SL}_2(\mathbb{Q}) \longrightarrow \mathrm{Aut}^{\otimes}(\omega),$$

$$\Phi(t, g)(V \otimes V_{\mathrm{PW}}) = \eta_V(t) \otimes \rho_{V_{\mathrm{PW}}}(g),$$

on the natural-substrate panel $\{V \otimes V_{\mathrm{PW}} :\ V \in \mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max})),\ V_{\mathrm{PW}} \in \mathfrak{PW}\}$*. At* $n_{\max} = 2$ *the* $\mathbb{G}_a$ *factor of* Φ *is injective on the full primitive panel.*

Remark 5.6 (Scope)**.** Corollary 5.5 closes the *inclusion direction* $\mathrm{Aut}^{\otimes}(\omega) \supseteq U^*_{\mathrm{Levi}}$ at finite cutoff with both Levi factors explicit. The full converse equality $\mathrm{Aut}^{\otimes}(\omega) = U^*_{\mathrm{Levi}}$ on *both* Levi factors is the Deligne–Milne 1982 Theorem 2.11 Tannakian reconstruction direction and would require a separate argument that no other natural $\otimes$-automorphisms of ω exist on the combined $n_{\max}$-axis $\times$ $j_{\max}$-axis substrate. The *abelian-factor* converse at finite cutoff is, however, closeable bit-exactly: this is the content of Section 5.5 below (Theorem 5.7). Full-panel injectivity at $n_{\max} \geq 3$ is sprint-scale (the construction is identical, only the panel size grows by $3N(n_{\max})$); extension of SL_2 to Peter–Weyl reps $\mathrm{Sym}^k V_{\mathrm{fund}}$ for $k \geq 3$ is likewise sprint-scale.

5.5 Finite-cutoff equality on the abelian factor (TC-2a)

We can close the converse equality on the *abelian factor* $\mathbb{G}_a^{3N(n_{\max})} \subseteq \mathrm{Aut}^{\otimes}(\omega)$ at finite cutoff bit-exactly. The construction is direct: parameterise every candidate natural $\otimes$-automorphism on the faithful witness panel by symbolic $\mathbb{Q}$-indeterminates and read off the dimension of the resulting bit-exact linear variety. This is the content of Sprint Q5'-Tannakian-Closure TC-2a, 2026-06-06; memo `debug/sprint_q5p_tc2a_aut_equality_memo.md`, data `debug/data/sprint_q5p_tc2a_aut_equalit` driver `debug/compute_q5p_tc2a_aut_equality.py`, tests `tests/test_tannakian_aut_equality.py`.

Theorem 5.7 (TC-2a, finite-cutoff equality on the abelian factor)**.** *At $n_{\max} = 2$, restrict the fiber functor ω to the $n_{\max}$-axis substrate $\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(2))$ (forgetting the Peter–Weyl decoration). The bit-exact $\mathbb{Q}$-linear variety of natural $\otimes$-automorphisms of ω on the faithful witness panel of Theorem 5.4 together with the unit object $\mathbf{1}$ has dimension*

$$\dim \mathrm{Aut}^{\otimes}(\omega)\big|_{n_{\max}=2,\ \text{abelian factor}} \;=\; 15 \;=\; 3N(2) \;=\; \dim \mathbb{G}_a^{3N(2)}(\mathbb{Q}).$$

Equivalently, the closed-form solution of the naturality system on the witness panel is

$$\eta_{V_g} = \begin{pmatrix} 1 & q_g \\ 0 & 1 \end{pmatrix} = \exp(q_g\, E_{12}) \qquad (\,q_g \in \mathbb{Q} \text{ free, one per primitive generator}), \qquad \eta_{\mathbf{1}} = 1,$$

matching $\Phi(t)(V_g)$ from Section 5.1 with $t_g = q_g$ bit-exactly on all 15 panel reps. The TC-1e/TC-1f inclusion $\mathbb{G}_a^{3N(2)} \subseteq \mathrm{Aut}^{\otimes}(\omega)$ is therefore an equality on the abelian factor at this cutoff.

Proof. Parameterise η_{V_g} on each of the 15 witness reps of Theorem 5.4 by four symbolic $\mathbb{Q}$-indeterminates (p_g, q_g, r_g, s_g), plus one $\eta_{\mathbf{1}}$ indeterminate for the unit object. Total: 61 indeterminates. For every ordered pair (V_a, V_b) in the panel extended by $\mathbf{1}$, compute the $\mathbb{Q}$-basis of $\mathrm{Hom}(V_a, V_b)$ by solving the intertwining linear system $f\, X_g^{V_a} = X_g^{V_b} f$ (for every primitive generator g) bit-exactly. The panel has 271 morphisms: 30 endomorphisms ($\dim \mathrm{End}(V_g) = 2$ for each g: centraliser of E_{12} in $M_2(\mathbb{Q})$), 210 cross-pair morphisms ($\dim \mathrm{Hom}(V_g, V_h) = 1$ for $g \neq h$: matrices of the form $\begin{pmatrix} 0 & a \\ 0 & 0 \end{pmatrix}$), 15 morphisms $\mathbf{1} \to V_g$ ($\dim = 1$: matrices $(a, 0)^T$), 15 morphisms $V_g \to \mathbf{1}$ ($\dim = 1$: matrices $(0, a)$), and the identity $\mathbf{1} \to \mathbf{1}$. Impose naturality $\eta_{V_b} f = f\, \eta_{V_a}$ for every basis morphism; at the matrix-entry level this produces 735 $\mathbb{Q}$-linear equations (dropping trivially-zero entries). Append the unit normalisation $\eta_{\mathbf{1}} = 1$ as the 736th equation. The resulting linear system $A\, v = b$ over $\mathbb{Q}$ has shape $A \in \mathbb{Q}^{736 \times 61}$, rank 46, and is consistent ($\mathrm{rank}([A \mid b]) = \mathrm{rank}(A) = 46$). The nullity is $61 - 46 = 15$. Solving in closed form: each panel rep solves to $p_g = 1, r_g = 0, s_g = 1$ with q_g free, and $\eta_{\mathbf{1}} = 1$. The 15 free parameters $\{q_g\}_g$ are in bijection with the 15 primitive generators of $\mathcal{H}_{\mathrm{GV}}(2)$. The closed-form $\eta_{V_g} = \exp(q_g\, E_{12}) = \Phi((\ldots, t_g = q_g, \ldots))(V_g)$ matches the TC-1e inclusion of Section 5.1 on all 15 reps bit-exact. □ □

Remark 5.8 (Scope of TC-2a)**.** Theorem 5.7 closes the converse direction on the *abelian factor* $\mathbb{G}_a^{3N(2)}$ at the single cutoff $n_{\max} = 2$. The SL_2 factor of U^*_{Levi} acts on the orthogonal Peter–Weyl decoration ($j_{\max}$ axis, Section 5.2) and is not detected by the $n_{\max}$-axis-only fiber functor used here; it requires the combined panel $V \otimes V_{\mathrm{PW}}$. The natural sprint-scale follow-on (TC-2b) layers in the Peter–Weyl panel of Theorem 5.2, parameterises $\eta_{V_g \otimes V_{\mathrm{PW}}}$ on combined reps, imposes naturality against the SL_2-intertwiners from Theorem 5.2 as well as the $\mathcal{H}_{\mathrm{GV}}$-intertwiners, and verifies $\dim = 18 = \dim U^*_{\mathrm{Levi}}$. Higher $n_{\max}$ extends both panels and re-verifies; the PS-2 cofiltered functoriality of Section 2.3 packages the per-cutoff verdicts into a pro-system coherence statement that, together with period-level coherence on $\mathcal{O}_\infty$ (PS-3) per v3.66.0 FO3 Interpretation C, would close the full long-range reconstruction. TC-2a is the first quantitative measurement of the converse direction; it is the abelian-factor sub-claim at $n_{\max} = 2$, not the full claim.

5.6 Finite-cutoff equality on the SL_2 factor (TC-2b)

The SL_2 factor of U^*_{Levi} is reconstructed at the Peter–Weyl panel level via a structurally distinct mechanism: naturality alone gives zero constraint on irreducible reps (Schur's lemma: $\mathrm{End}(V) = \mathbb{Q} \cdot I$ on each irreducible), so the binding constraint comes from *tensor compatibility* on $V_{\mathrm{fund}} \otimes V_{\mathrm{fund}}$ combined with its decomposition isomorphism $V_{\mathrm{fund}} \otimes V_{\mathrm{fund}} \cong \mathrm{Sym}^2 V_{\mathrm{fund}} \oplus V_{\mathrm{triv}}$. Sprint Q5'-Tannakian-Closure TC-2b, 2026-06-06; memo `debug/sprint_q5p_tc2b_sl2_aut_equality_memo.md`, data `debug/data/sprint_q5p_tc2b_sl2_aut_equality.json`, driver `debug/compute_q5p_tc2b_sl2_aut_equa` tests `tests/test_tannakian_sl2_aut_equality.py`.

Theorem 5.9 (TC-2b, finite-cutoff equality on the SL_2 factor)**.** *On the Peter–Weyl panel* $\{V_{\mathrm{triv}}, V_{\mathrm{fund}}, \mathrm{Sym}^2 V_{\mathrm{fund}}\}$ *with the explicit* $\mathbb{Q}$*-linear decomposition isomorphism* $\Phi : V_{\mathrm{fund}} \otimes V_{\mathrm{fund}} \xrightarrow{\sim} \mathrm{Sym}^2 V_{\mathrm{fund}} \oplus V_{\mathrm{triv}}$, $\dim \mathrm{Aut}^{\otimes}(\omega)|_{\mathfrak{PW}} = 3 = \dim SL_2$*. Tensor compatibility produces the bit-exact identity*

$$\Phi \cdot (\eta_{V_{\mathrm{fund}}} \otimes \eta_{V_{\mathrm{fund}}}) \cdot \Phi^{-1} = \mathrm{Sym}^2(\eta_{V_{\mathrm{fund}}}) \oplus \det(\eta_{V_{\mathrm{fund}}}),$$

which forces $\eta_{\mathrm{Sym}^2} = \mathrm{Sym}^2(\eta_{V_{\mathrm{fund}}})$ *(top-left* 3×3 *block) and* $\eta_{V_{\mathrm{triv}}} = \det(\eta_{V_{\mathrm{fund}}})$ *(bottom-right* 1×1 *block). Unit normalisation* $\eta_{V_{\mathrm{triv}}} = 1$ *then forces* $\det(\eta_{V_{\mathrm{fund}}}) = 1$*, i.e.* $\eta_{V_{\mathrm{fund}}} \in SL_2(\mathbb{Q})$*.*

Proof. Parameterise $\eta_{V_{\mathrm{fund}}} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ symbolically over $\mathbb{Q}$. The standard Sym^2 formula (`geovac.tannakian._sl` used in TC-1f Theorem 5.2) gives $\mathrm{Sym}^2(\eta_{V_{\mathrm{fund}}})$ as the 3×3 matrix

$$\mathrm{Sym}^2(\eta_{V_{\mathrm{fund}}}) = \begin{pmatrix} a^2 & 2ab & b^2 \\ ac & ad + bc & bd \\ c^2 & 2cd & d^2 \end{pmatrix}.$$

The decomposition isomorphism in basis $\{e_1 \otimes e_1, e_1 \otimes e_2, e_2 \otimes e_1, e_2 \otimes e_2\}$ (source) and $\{e_1^2, e_1 e_2, e_2^2, e_1 \wedge e_2\}$ (target) is the explicit 4×4 rational matrix

$$\Phi = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1/2 & -1/2 & 0 \end{pmatrix}, \qquad \Phi^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

`sympy`-symbolic computation gives

$$\Phi \cdot (\eta_{V_{\mathrm{fund}}} \otimes \eta_{V_{\mathrm{fund}}}) \cdot \Phi^{-1} = \begin{pmatrix} a^2 & 2ab & b^2 & 0 \\ ac & ad + bc & bd & 0 \\ c^2 & 2cd & d^2 & 0 \\ 0 & 0 & 0 & ad - bc \end{pmatrix},$$

in which the top-left 3×3 block is precisely $\mathrm{Sym}^2(\eta_{V_{\mathrm{fund}}})$ and the bottom-right 1×1 block is $\det(\eta_{V_{\mathrm{fund}}}) = ad - bc$; off-diagonal blocks vanish identically. With unit normalisation $\eta_{V_{\mathrm{triv}}} = 1$, the bottom-right block forces $ad - bc - 1 = 0$, a single polynomial constraint on the 4-dim ambient $M_2(\mathbb{Q})$. The Jacobian $\nabla(ad - bc - 1) = (d, -c, -b, a)$ has rank 1 at every SL_2-point (verified at $\eta = I$: $(1, 0, 0, 1)$, rank 1; at generic $\eta = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$: $(1, -1, -1, 2)$, rank 1). The variety has codim 1, $\dim = 4 - 1 = 3 = \dim SL_2$. □ □

Corollary 5.10 (TC-2a + TC-2b). *At $n_{\max} = 2$ on the combined natural substrate (the $n_{\max}$-axis $\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(2))$ tensored with the Peter–Weyl panel), $\dim \mathrm{Aut}^{\otimes}(\omega)|_{\mathrm{combined}} = 15+3 = 18 = \dim U^*_{\mathrm{Levi}}$. The TC-1e/TC-1f inclusion $U^*_{\mathrm{Levi}} \subseteq \mathrm{Aut}^{\otimes}(\omega)$ is therefore an equality at finite cutoff.*

Proof. By the exterior-tensor-product theorem for neutral Tannakian categories [2], for two neutral Tannakian categories C_1, C_2 over $\mathbb{Q}$ with fiber functors ω_1, ω_2, $\mathrm{Aut}^{\otimes}(\omega_1 \boxtimes \omega_2) = \mathrm{Aut}^{\otimes}(\omega_1) \times \mathrm{Aut}^{\otimes}(\omega_2)$. Applying this to $C_1 = \mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(2))$ with ω_1 the TC-1d no-op forgetful, and C_2 the PW sub-category of $\mathrm{Rep}(SL_2)$ generated by $\{V_{\mathrm{triv}}, V_{\mathrm{fund}}, \mathrm{Sym}^2\}$ with ω_2 the analogous forgetful, the per-factor dims of Theorems 5.7 and 5.9 give $\dim(15) + \dim(3) = 18 = \dim U^*_{\mathrm{Levi}}$. □

Provenance caveat. This corollary's only external warrant is the exterior-tensor-product theorem for neutral Tannakian categories. We cite Deligne's *Catégories tannakiennes* for it, but have not been able to verify a specific numbered statement against the primary text; the section pointer previously given here is withdrawn as unverified. The factorization $\mathrm{Aut}^{\otimes}(\omega_1 \boxtimes \omega_2) = \mathrm{Aut}^{\otimes}(\omega_1) \times \mathrm{Aut}^{\otimes}(\omega_2)$ is standard, but a reader wanting the citable form should consult the primary source directly. □

5.7 Higher cutoffs and the $n_{\max}$-uniform pattern (TC-2c)

The TC-2a abelian-factor argument is parametric in $n_{\max}$: the witness panel grows by $3N(n_{\max})$ generators at each cutoff, the constraint count grows polynomially, and the rank-nullity pattern is $n_{\max}$-uniform. Sprint Q5'-Tannakian-Closure TC-2c, 2026-06-06; memo `debug/sprint_q5p_tc2c_higher_cuto` driver `debug/compute_q5p_tc2c_higher_cutoff.py`, tests `tests/test_tannakian_higher_cutoff.py`.

Theorem 5.11 (TC-2c, finite-cutoff equality at higher cutoffs). *For every $n_{\max} \in \{1, 2, 3, 4\}$, $\dim \mathrm{Aut}^{\otimes}(\omega)|_{n_{\max}\text{-axis}} = 3N(n_{\max})$ bit-exactly, with the closed-form solution $\eta_{V_g} = \exp(q_g\, E_{12})$ on every panel rep matching $\Phi(t)(V_g)$ from Section 5.1 via the bijection $t_g = q_g$.*

Proof (parametric extension of Theorem 5.7). The argument of Theorem 5.7 is uniform in $n_{\max}$: the witness panel $\{V_g\}_{g \in P^{(n_{\max})}}$ scales with $|P^{(n_{\max})}| = 3N(n_{\max})$, the parameterisation gives $3N(n_{\max}) \cdot 4 + 1 = 12N(n_{\max}) + 1$ symbolic indeterminates, naturality on the morphism panel reduces the system to nullity $3N(n_{\max})$ via the same closed-form argument (every η_{V_g} solves to unipotent upper triangular with one free parameter q_g; $\eta_T = 1$ falls out from the unit pair). The bit-exact `sympy`-rational verification at $n_{\max} \in \{1, 2, 3, 4\}$:

$n_{\max}$	$3N(n_{\max})$	morphisms	constraints	nullity	wall (s)
1	6	49	132	**6**	~ 1
2	15	271	736	**15**	1.85
3	27	811	2,296	**27**	12.23
4	42	1,891	5,461	**42**	55.74

Wall time grows polynomially in $n_{\max}$; the structural pattern is not isolated to the smallest nontrivial cutoff but $n_{\max}$-uniform. Φ-recovery is $3N(n_{\max})/3N(n_{\max})$ bit-exact at every cutoff. □ □

Remark 5.12 (Rank-nullity closed form). The rank of the naturality + unit constraint matrix is $\mathrm{rank}(A) = 3 \cdot 3N(n_{\max}) + 1 = 9N(n_{\max}) + 1$ exactly (verified values: 19, 46, 82, 127; equivalently forced by rank–nullity from the $3N$ nullity recorded in Theorem 5.7's proof, which gives $\mathrm{rank}(A) = (12N + 1) - 3N$). This matches the structural picture: naturality forces $p_g = 1$,

$r_g = 0$, $s_g = 1$ for each g (three linear constraints per generator) plus the unit normalisation $\eta_T = 1$, totalling $9N(n_{\max}) + 1$ *symbol fixings* on the $12N(n_{\max}) + 1$ indeterminates. All of them are independent over the morphism panel data, so the solution space has dimension $(12N + 1) - (9N + 1) = 3N$—which is the rank pattern above, by rank–nullity. The closed-form generalisation across all $n_{\max} \geq 1$ is conjectured and likely sprint-scale to verify symbolically; TC-2c does not pursue this sharpening.

5.8 Pro-system cofiltered coherence (TC-2d)

The per-cutoff equalities of Theorems 5.7, 5.9, and 5.11 package into a single pro-system statement via the PS-2 cofiltered functoriality of Section 2.3. Sprint Q5'-Tannakian-Closure TC-2d, 2026-06-06; memo `debug/sprint_q5p_tc2d_cofiltered_coherence_memo.md`, driver `debug/compute_q5p_tc2` tests `tests/test_tannakian_cofiltered_coherence.py`.

Theorem 5.13 (TC-2d, cofiltered coherence on $\mathrm{Aut}^{\otimes}(\omega)$)**.** *For every pair $k \leq m$, the Spec-dual of the PS-2 Hopf-algebra transition $\Phi_{m,k} : \mathcal{H}_{\mathrm{GV}}(m) \to \mathcal{H}_{\mathrm{GV}}(k)$ is a restriction map*

$$\rho_{m,k} : \ \mathrm{Aut}^{\otimes}(\omega)^{(m)} \to \mathrm{Aut}^{\otimes}(\omega)^{(k)},$$

which acts at the parameter level as coordinate truncation $t^{(m)} \mapsto t^{(m)}\big|_{P^{(k)}}$ on $\mathbb{Q}^{3N(m)}$. The restriction is a group homomorphism, satisfies the cofiltered transitivity $\rho_{n,k} = \rho_{m,k} \circ \rho_{n,m}$ at every triple $k \leq m \leq n$, and is bit-exact on every witness rep V_g for $g \in P^{(k)} \subseteq P^{(m)}$.

Proof. At the parameter level the restriction is the obvious dict truncation. On a witness rep V_g with $g \in P^{(k)}$, $\Phi(t^{(m)})(V_g) = \exp(t_g^{(m)} E_{12})$ (since $X_h^{V_g} = 0$ for $h \neq g$) is the same as $\Phi(t^{(m)}\big|_{P^{(k)}})(V_g)$ because the g-component of $t^{(m)}\big|_{P^{(k)}}$ equals the g-component of $t^{(m)}$. Group-hom property follows from $\Phi(t_1)\Phi(t_2) = \Phi(t_1 + t_2)$ (abelian unipotent factor) and $\mathbb{Q}$-linearity of parameter truncation. Transitivity at the parameter level reduces to $t|_{P^{(k)}} = (t|_{P^{(m)}})|_{P^{(k)}}$, a trivial set-theoretic identity for $P^{(k)} \subseteq P^{(m)} \subseteq P^{(n)}$.

The bit-exact verification: Block A (panel-level restriction coherence on V_g) tested at 6 pairs $(k, m) \in \{(1, 2), (1, 3), (2, 3), (1, 4), (2, 4), (3, 4)\}$ across 3 test parameter seeds each, totalling 225 residuals; Block B (transitivity at 4 triples $\times 3$ seeds) totalling 12; Block C (group-hom at 3 pairs $\times 2$ generators each) totalling 6. All 243 residuals are bit-exact zero. □ □

Corollary 5.14 (Inverse-limit Tannakian dual at the panel level)**.** *The cofiltered system $\{\mathrm{Aut}^{\otimes}(\omega)^{(n)}\}_{n \in \mathbb{N}}$ with restriction maps $\{\rho_{m,k}\}_{k \leq m}$ has inverse limit*

$$\mathrm{Aut}^{\otimes}(\omega)^{(\infty)} = \varprojlim_n \mathrm{Aut}^{\otimes}(\omega)^{(n)} = \varprojlim_n \big(\mathbb{G}_a^{3N(n)} \rtimes SL_2\big) = \mathbb{G}_a^{\infty} \rtimes SL_2,$$

at the natural-substrate panel level. The remaining content of the full long-range reconstruction is the pro-finite Tannakian theorem [2, 3] coherent with v3.66.0 FO3 Interpretation C at the period level on $\mathcal{O}_\infty$ (PS-3) of Section 2.4 — a structural identification between $\mathbb{G}_a^{\infty} \rtimes SL_2$ and the motivic Galois group on the mixed-Tate period ring $F(s)$ of $\mathcal{O}_\infty$.

6 The level-4 cosmic-Galois injection

The per-cutoff equality of Corollary 5.14 identifies the Tannakian dual $\mathrm{Aut}^{\otimes}(\omega)^{(\infty)}$ with $\mathbb{G}_a^{\infty} \rtimes SL_2$ as a pro-algebraic group, but does not yet compare it to a published motivic-Galois target. This section closes that comparison in the *injection direction*: we exhibit an explicit homomorphism of pro-algebraic groups $\Phi^{\mathrm{inj}} : U_{GV}^* = \mathbb{G}_a^{\infty} \rtimes SL_2 \to \mathcal{U}_4 \rtimes SL_2$ onto the *abelianized* level-4 image (Reading A; not a closed immersion — see Theorem 6.2), where $\mathcal{U}_4$ is the pro-unipotent radical of the level-4 motivic Galois group $\mathcal{G}_4 = \mathcal{G}_{\mathrm{MT}(\mathbb{Z}[i,1/2])}$ of Deligne 2010 [7] and Glanois 2015 [8]. The four pieces of the construction — the period-level map, its multiplicativity, its coproduct compatibility, and the SL_2 factor's behaviour under the level-4 cyclotomic character χ_4 — are all in the literature; the assembly is the work of this section.

6.1 The period map

The level-4 motivic Galois group has Levi decomposition

$$\mathcal{G}_4 \;=\; \mathbb{G}_m \ltimes \mathcal{U}_4,$$

where $\mathbb{G}_m$ acts on each weight piece by the cyclotomic character χ_4 and $\mathcal{U}_4$ is the pro-unipotent radical generated by depth-graded motivic MZVs $\{\zeta^{\mathfrak{m}}(s_1,\ldots,s_k)\}$ at level $N=4$ (Deligne 2010 [7]; Glanois 2015 explicit basis $\mathcal{B}^4$ [8]). The natural action of $\mathcal{G}_4$ on periods is faithful at level 4. This is not a general fact and we do not state it as one: faithfulness holds only in the exceptional cases $N \in \{2,3,4,6,8\}$ (Deligne 2010 [7], §4 *Fidelité*), together with $N=1$ (Brown 2012), and it *fails* for prime $N \geq 5$, where Goncharov showed the action is already unfaithful on $\wedge^2\mathfrak{u}^1_\omega$. Level 4 is one of the exceptional cases, which is all this paper needs; the underlying mixed-Tate package is Deligne–Goncharov [26]. So a *period-content map* $\mathcal{H}_{\mathrm{GV}} \to \mathrm{Periods}(\mathcal{G}_4)$ that respects the GeoVac Hopf-algebra structure dualises to a homomorphism $U^*_{GV} \to \mathcal{G}_4$ of pro-algebraic groups.

Concretely, the period map evaluates GeoVac's primitive Hopf generators on their structural period content (Paper 55).

Definition 6.1. The *level-4 period map*

$$\pi:\ \mathcal{H}_{\mathrm{GV}} \longrightarrow \mathrm{MT}(\mathbb{Z}[i,1/2])$$

sends a primitive generator $x_{(n,l),k}$ of $\mathcal{H}_{\mathrm{GV}}$ to its image under the master Mellin engine evaluator $\mathcal{M}[\mathrm{Tr}(D^k\, e^{-tD^2})]$ at the Camporesi–Higuchi spectrum on S^3 (Paper 32 §VIII; Paper 18 §III.7):

$$\pi(x_{(n,l),0}) \;=\; m_1^{(n,l)} \in M_1 \;\subset\; \mathbb{Q}[\pi,\pi^{-1}],$$

$$\pi(x_{(n,l),2}) \;=\; m_2^{(n,l)} \in M_2 \;\subset\; \bigoplus_{j\geq 0} \pi^{2j}\cdot\mathbb{Q},$$

$$\pi(x_{(n,l),1}) \;=\; m_3^{(n,l)} \in M_3 \;\subset\; \mathrm{MT}(\mathbb{Z}[i,1/2]),$$

where the $k=1$ (M3) evaluation carries the sector's *full* first-order period content — both the ungraded trace and the proven χ_{-4} vertex-parity grading $\mathrm{Tr}((-1)^N D\, e^{-tD^2})$ (Paper 28 [28] Theorem 3), so $m_3^{(n,l)}$ records which quarter-integer Hurwitz shift-class the sector feeds (3/4 if its Camporesi–Higuchi shell is even, 5/4 if odd; Remark 6.6). The map is extended multiplicatively to all of $\mathcal{H}_{\mathrm{GV}} = \mathrm{Sym}_{\mathbb{Q}}(V)$ (Paper 55 Theorems 3.2, 4.1, and 5.1 establish the three target inclusions).

The three target rings M_1, M_2, M_3 sit inside a common ambient cyclotomic mixed-Tate ring at level ≤ 4 over $\mathbb{Z}[i,1/2]$ (Paper 55 §6 role-disjointness lemma); this is the period content of $\mathcal{G}_4$ at the levels reachable by GeoVac's natural three-bullet partition.

6.2 The injection theorem

Theorem 6.2 (Level-4 cosmic-Galois injection)**.** *The period map π of Definition 6.1 dualises to a homomorphism of pro-algebraic groups*

$$\Phi^{\mathrm{inj}}:\ U^*_{GV} = \mathbb{G}_a^\infty \rtimes SL_2 \;\longrightarrow\; \mathcal{G}_4 = \mathbb{G}_m \ltimes \mathcal{U}_4,$$

landing in the level-4 cosmic-Galois group. Under the per-sector period evaluation of Definition 6.1, the M1 and M2 columns each collapse to a single rank-1 target per (Mellin slot, Tate weight), while the M3 column has rank 2 — the two χ_{-4} vertex-parity classes (Remark 6.6); its parity-blind restriction (the per-sector η-content alone; §6.3, `tests/test_paper56_injection_g4_periodmap.p` *is rank-1. In every case the rank is far below the sector count N, so Φ^{inj} is* not *injective on the sector-indexed generators; it factors through the abelianization $U^*_{GV} \twoheadrightarrow (U^*_{GV})^{\mathrm{ab}}$ and realises the level-4* abelianized *image (Reading A, Remark 6.5). The earlier "closed-immersion" reading required* full *M3-column injectivity (all N distinct generators mapping to $\mathbb{Q}$-linearly independent periods); the genuine Gram computation refutes it (*$\mathrm{rank} = 2 \ll N$*).*

The four structural compatibilities are:

1. ***(C1) Multiplicativity.*** *$\pi(xy) = \pi(x)\,\pi(y)$ for every $x, y \in \mathcal{H}_{\mathrm{GV}}$, dualising to Φ^{inj} being a group homomorphism.*

2. ***(C2) Coproduct compatibility (depth-1).*** *The GeoVac primitive coproduct $\Delta x = x \otimes 1 + 1 \otimes x$ maps to the depth-1 part of the motivic coproduct on $\mathcal{O}(\mathcal{G}_4)$: for every primitive generator x, $(\pi \otimes \pi)(\Delta x) = \Delta^{\mathrm{mot}}_{\mathrm{depth}=1}\pi(x)$ in $\mathcal{O}(\mathcal{G}_4) \otimes \mathcal{O}(\mathcal{G}_4)$.*

3. ***(C3) SL_2-to-Levi via cyclotomic character.*** *GeoVac's SL_2 factor (acting on the Peter–Weyl decoration V_{PW}) maps to $\mathcal{G}_4$ via the trivial composition $SL_2 \xrightarrow{\det \equiv 1} \mathbb{G}_m \hookrightarrow \mathcal{G}_4$ on the reductive factor. Equivalently, the SL_2-content of U^*_{GV} sits entirely inside the unipotent radical $\mathcal{U}_4$ at the level of periods (the $\mathbb{G}_m$-weight grading is fixed under $\Phi^{\mathrm{inj}}(SL_2)$). The non-trivial SL_2 action on $\mathrm{Sym}^k V_{\mathrm{fund}}$ is detected at the representation level (Theorem 5.2), not at the period-grading level.*

4. ***(C4) Abelianized image (Reading A), not a closed immersion.*** *Under the enriched per-sector period content (Definition 6.1) the M3 column has rank 2 per weight — the two χ_{-4} vertex-parity classes (Remark 6.6) — while M1 and M2 collapse to rank-1; every rank is $\ll N$, so Φ^{inj} is not injective on the sector-indexed generators and factors through the abelianization. The well-defined object is the abelianized image $\Phi^{\mathrm{inj}}((U^*_{GV})^{\mathrm{ab}})$, cut out inside $\mathcal{G}_4$ by the Glanois 2015 basis relations [8] on the depth-graded MZV generators that fail to appear in the master Mellin engine output. At each weight $w \geq 1$, the codimension of the (abelianized) image in $\mathcal{U}_4$ is the number of level-4 depth-≥ 2 motivic MZVs at weight w not realised by any GeoVac M3 observable, positive at depth ≥ 2 (see honest scope below).*

Proof. (C1) Multiplicativity. The GeoVac Hopf algebra $\mathcal{H}_{\mathrm{GV}} = \mathrm{Sym}_{\mathbb{Q}}(V)$ is the free commutative algebra on the primitive space V. Period evaluation π is defined on generators (Definition 6.1) and extended multiplicatively to all of $\mathcal{H}_{\mathrm{GV}}$ by construction. By the universal property of $\mathrm{Sym}_{\mathbb{Q}}(V)$, this extension is unique and multiplicative: $\pi(xy) = \pi(x)\,\pi(y)$ holds bit-exact for every $x, y \in \mathcal{H}_{\mathrm{GV}}$. The dual map $\Phi^{\mathrm{inj},*} : \mathcal{O}(\mathcal{G}_4) \to \mathcal{O}(U^*_{GV})$ is then a $\mathbb{Q}$-algebra homomorphism (standard Hopf-algebra duality), so Φ^{inj} is a group homomorphism.

(C2) Coproduct compatibility at depth 1. On a connected graded Hopf algebra the depth-1 (lowest coradical-filtration) graded piece coincides with the space of primitives; on the *abelianization* this is the Cartier–Milnor–Moore identification of indecomposables with primitives [27]. On each M_i column, the depth-1 part of $\Delta^{\mathrm{mot}}_{\mathrm{depth}=1}\pi(x_{(n,l),k})$ is exactly $\pi(x_{(n,l),k}) \otimes 1 + 1 \otimes \pi(x_{(n,l),k})$ because (i) M1 and M2 outputs are depth-0 (pure-Tate even-weight, Paper 55 Theorem 4.1) so they are central in the coproduct, and (ii) M3 outputs at depth 1 (single Hurwitz/Dirichlet-L values) are primitive in the depth-1 sub-quotient by that coradical identification. Higher-depth content is not detected by GeoVac's primitive coproduct; this is the Reading-A abelianization noted in Section 8.3.

(C3) SL_2-to-Levi via χ_4. The cyclotomic character $\chi_4 : \mathcal{G}_4 \to \mathbb{G}_m$ acts on each weight-w period piece as scaling by $\chi_4(g) = g^w$ (Deligne 2010 [7] §4). GeoVac's SL_2 acts on the Peter–Weyl decoration via the standard Sym^k action (Theorem 5.2); the composition $SL_2 \xrightarrow{\det} \mathbb{G}_m$ is trivial because $\det(g) = 1$ for all $g \in SL_2(\mathbb{Q})$. Hence $\chi_4 \circ \Phi^{\mathrm{inj}}|_{SL_2} \equiv 1$, i.e. the SL_2-image preserves the $\mathbb{G}_m$-weight grading. By the Levi decomposition $\mathcal{G}_4 = \mathbb{G}_m \ltimes \mathcal{U}_4$, an element acting trivially on the reductive factor sits in the unipotent radical $\mathcal{U}_4$ (or its commutant); since SL_2

also acts non-trivially via Sym^k, the image is genuinely inside $\mathcal{U}_4 \rtimes SL_2$ where the semidirect product structure is induced by the natural action of SL_2 on the Sym^k decomposition of the depth-1 motivic Lie algebra.

(C4) Abelianized image via Glanois basis (not a closed immersion). The Glanois 2015 explicit basis $\mathcal{B}^4$ [8] stratifies $\mathrm{Lie}(\mathcal{U}_4)$ by depth and weight. GeoVac's image at weight w and depth $d = 1$ is spanned by the M3 column generators $\{\pi(x_{(n,l),1})\}$. The Goncharov–Deligne 2005 faithfulness result [26] would ensure that distinct M3 generators map to $\mathbb{Q}$-linearly independent periods *provided* the per-sector M3 evaluation were rank ≥ 2. However, the only concrete per-sector M3 content the corpus supplies is the single scalar $\eta_{(n,l)}$ (Paper 55), so the parity-blind restriction has all period vectors collinear: its Gram is rank 1 (`tests/test_paper56_injection_g4_periodmap.py`). The enriched map (Definition 6.1) adds the χ_{-4} vertex-parity grading, which sorts sectors into the two shift-classes 3/4 (even shell) and 5/4 (odd shell); its Gram has rank 2 (Remark 6.6, `tests/test_paper56_m3_parity_rank.py`). Because $2 \ll N$, Φ^{inj} is still *not* injective on the sector-indexed generators. What is well-defined is the abelianized image $\Phi^{\mathrm{inj}}((U^*_{GV})^{\mathrm{ab}})$, Zariski-closed in $\mathcal{U}_4$ (the kernel of the finitely many linear functionals vanishing on the realised M3 content, each finite-weight piece finite-dimensional over $\mathbb{Q}$ by Glanois 2015). A genuinely injective (closed-immersion) refinement would require all N sectors to map to $\mathbb{Q}$-linearly independent periods; the χ_{-4} vertex-parity route is capped at rank 2 (two parity classes), so it refines the rank-1 collapse but does *not* reach a closed immersion for $N > 2$ (Remark 6.6). □ □

Remark 6.3 (Why level 4 and not level 1)**.** The refinement to $\mathcal{G}_4$ (rather than Brown's $\mathcal{G}_{\mathrm{MT}(\mathbb{Z})}$ at level 1) is motivated by Eskandari–Murty–Nemoto 2025 [25]: Catalan $G = \beta(2)$ *is* mixed Tate over $\mathbb{Q}(i) = \mathbb{Q}[i, 1/2]^{\times\otimes\mathbb{Q}}$ at level 4. *Honest scope:* the converse—that G is not a period of mixed Tate motives over $\mathbb{Q}$—is expected but, to our knowledge, open; the level-4 choice therefore rests on where G is *known* to live, not on a proven exclusion at level 1. Since GeoVac's M3 vertex-parity sector produces G in QED vertex sums (Paper 28 Theorem 3, Sub-sector 2), the natural cosmic-Galois target must contain at least level 4. No level ≥ 5 is needed: Paper 55 §5 shows that every M3 output on Camporesi–Higuchi S^3 is a Hurwitz value at quarter-integer shift $\{1/4, 3/4, 5/4\}$, which the explicit Glanois 2015 basis realises inside $\mathrm{MT}(\mathbb{Z}[i, 1/2])$.

Remark 6.4 (Reading-A abelianization at depth 1)**.** The injection of Theorem 6.2 sees only the *abelianization* of $\mathcal{U}_4$: the image $\Phi^{\mathrm{inj}}(\mathbb{G}_a^{\infty})$ sits inside the depth-1 part $\mathrm{gr}^1_D\mathcal{U}_4$, since the GeoVac primitive coproduct $\Delta x = x \otimes 1 + 1 \otimes x$ pulls back the motivic coproduct only at depth 1 (proof of (C2) above). This matches the NA-1 structural finding (Section 8.3, Reading A): GeoVac's primitive Hopf substrate gives an abelian unipotent radical by the Cartier–Milnor–Moore theorem ([27] as recalled in Brown 2012 [3] §5), so the natural Tannakian dual $\mathbb{G}_a^{\infty}$ is the abelianization of any free non-abelian unipotent target. The depth-1 sub-quotient of $\mathcal{U}_4$ is exactly the abelianization $\mathcal{U}_4^{\mathrm{ab}} \cong \mathrm{gr}^1_D\mathcal{U}_4$, and this is what Φ^{inj} sees on the unipotent side.

Remark 6.5 (Honest scope — partiality)**.** Three honest caveats refine the partial nature of the injection (Track C scoping memo, end-of-day 2026-06-06 synthesis):

- *M1 and M2 column collapse.* On the M1 column the period ring is $\mathbb{Q}[\pi, \pi^{-1}]$, 1-dimensional per primitive generator at any fixed Tate weight; on the M2 column the period ring is $\bigoplus_k \pi^{2k}\mathbb{Q}$, 1-dimensional per even Tate weight. The GeoVac side has up to $N(n_{\max})$ distinct primitive generators per (Mellin slot, Tate weight) pair (one per Coulomb sector (n, l)), but the period side has only one. The map $\Phi^{\mathrm{inj}}|_{M_1}$ and $\Phi^{\mathrm{inj}}|_{M_2}$ each factor through a quotient that collapses the multiple sector-indexed generators to a single per-weight target. This is not a failure of the construction — it is the structural manifestation of GeoVac's three-bullet master Mellin engine partition (Paper 18 §III.7), where M1 and M2 are pure-Tate and sector-blind by design.
- *The M3 column is rank 2 (parity-blind restriction rank-1).* The M3 column ($k = 1$, vertex-parity Hurwitz) is the only column mapping to a depth-graded target with non-trivial

cyclotomic content. Its parity-blind per-sector content is the single scalar $\eta_{(n,l)}$ (Paper 55), under which each sector's period vector is $\eta_{(n,l)}$ times a shared Hurwitz block — collinear, Gram rank 1, det 0 (`tests/test_paper56_injection_g4_periodmap.py`). The faithful Goncharov–Deligne 2005 action on $\pi_1^{\mathrm{mot}}(\mathbb{G}_m - \mu_4)$ would give *full* injectivity only for a rank-N M3 period map. The *enriched* map (Definition 6.1) adds the proven χ_{-4} vertex parity — the per-(n,l) parity decomposition of $D(s)$ — which sorts sectors into the two shift-classes $3/4, 5/4$ and raises the M3 rank to 2 (Remark 6.6, `tests/test_paper56_m3_parity_rank.py`), capped there. Since $2 \ll N$, Φ^{inj} still realises the *abelianized* image (Reading A) on all three columns.

- *Depth-blindness on the GeoVac side.* The injection is constructed at depth 1 (linear combinations of single Hurwitz values). GeoVac's primitive Hopf algebra cannot detect shuffle / deconcatenation structure (Reading A, Section 8.3), so the map cannot be promoted to a depth-respecting injection without enriching the substrate to the cofree-cocommutative shuffle Hopf algebra $T(V)$. Whether such an enrichment is structurally required (Reading B wins NA-1 sprint) or whether the abelianization is the correct geometric realisation (Reading A wins) is the genuinely open question pending the sprint-scale NA-1 depth-2 Mellin test.

Remark 6.6 (Vertex-parity resolution of the M3 column)**.** The M3 column's parity-*blind* restriction is rank-1 — its per-sector content is the single scalar $\eta_{(n,l)}$ (Remark 6.5). This remark backs the rank-2 statement of Theorem 6.2(C4). The follow-on — a "sector-resolved rank-≥ 2 M3 period map, a per-(n,l) parity decomposition of $D(s)$" — is supplied by the *proven* χ_{-4} vertex-parity identity (Paper 28 [28] Theorem 3): even Camporesi–Higuchi shells contribute at Hurwitz shift 3/4 and odd shells at 5/4, with $D_{\mathrm{even}}(s) - D_{\mathrm{odd}}(s) = 2^{s-1}(\beta(s) - \beta(s-2))$ pure level-4 Dirichlet-L content. Each Coulomb sector (n,l) sits in its shell's parity class, so the parity-resolved period vector is $\eta_{(n,l)}$ times the basis vector of its shift-class, and the Gram rank equals the number of populated shift-classes: **rank 2 at every** $n_{\max} \geq 2$ (rank 1 at $n_{\max} = 1$, where only odd shells exist), bit-exact under both sector$\leftrightarrow$shell offset conventions (`tests/test_paper56_m3_parity_rank.py`). The two shift-classes are a genuine independent period, not a manufactured sign: $D_{\mathrm{even}}(4)$ and $D_{\mathrm{odd}}(4)$ are $\mathbb{Q}$-linearly independent in $\{\pi^2, \pi^4, G, \beta(4)\}$.

Two consequences. First, this is why Theorem 6.2(C4) records rank 2 (not the parity-blind rank-1): the M3 image is 2-dimensional — the two shift-classes $\{3/4, 5/4\}$, equivalently the two motivic *levels* (level-2 un-restricted $D(s) \oplus$ level-4 χ_{-4} difference) in the sum/difference basis. Second — sharpening the follow-on's implicit hope that "rank-$\geq 2 \Rightarrow$ injectivity" — the parity route is *capped* at rank 2 (there are only two parity classes), so it *cannot* deliver a faithful closed immersion (rank N) for $N > 2$: GeoVac is a strict, *rank-2* abelianized subgroup, neither rank-1 nor faithful.

Scope. The χ_{-4}-graded trace $\mathrm{Tr}((-1)^N D\, e^{-tD^2})$ is admitted into the period map (Definition 6.1) as genuine first-order period content — it is the same $k = 1$ Mellin slot and a proven GeoVac observable (Paper 28 [28] Theorem 3). Under this enriched map the M3 column is rank 2, which is the rank recorded by Theorem 6.2(C4); the parity-blind restriction to the ungraded trace alone is the rank-1 sub-statement.

6.3 Numerical verification panel

The four compatibilities (C1)–(C4) admit explicit bit-exact verifications at every finite cutoff $n_{\max} \in \{1, 2, 3, 4\}$. The driver, data, memo, and regression tests are:

- Driver: `tests/test_paper56_injection_g4.py`.
- Memos: `debug/sprint_injection_g4_memo.md` (foundational sprint, $n_{\max} = 2$); `debug/sprint_injectio` (this section's extension to $n_{\max} \in \{3, 4\}$).

The four panels have closed-form per-cutoff residual counts that depend only on $N(n_{\max}) = n_{\max}(n_{\max}+3)/2$:

- *(C1) Multiplicativity panel.* For every ordered pair (x, y) of primitive generators of $\mathcal{H}_{\mathrm{GV}}(n_{\max})$, the formal multiplicativity $\pi(xy) = \pi(x)\,\pi(y)$ holds in $\mathrm{Sym}_{\mathbb{Q}}(V_{n_{\max}})$ by the universal property of the symmetric algebra. Closed form: $(3N(n_{\max}))^2$ bit-exact zero residuals.

- *(C2) Coproduct panel.* For every primitive generator x in $\mathcal{H}_{\mathrm{GV}}(n_{\max})$, the equality $\Delta x = x \otimes 1 + 1 \otimes x$ at depth 1 is bit-exact by definition of the primitive coproduct. Closed form: $3N(n_{\max})$ bit-exact zero residuals.

- *(C3) SL_2-to-Levi panel.* For every g in the five-element $SL_2(\mathbb{Q})$ panel of Theorem 5.2 (identity, unipotent, torus, generic, Weyl), $\det(g) = 1$ bit-exact. The SL_2-to-Levi check is $n_{\max}$-independent because the cyclotomic character χ_4 acts on the reductive factor and is blind to the $n_{\max}$-axis substrate; closed form 5 bit-exact zero residuals per cutoff.

- *(C4) M3 period-map panel (abelianized; injectivity refuted).* For each of the $N(n_{\max})$ M3-column generators of $\mathcal{H}_{\mathrm{GV}}(n_{\max})$ (the $k = 1$ Mellin slot), the period-side image is the single per-sector scalar $\eta_{(n,l)}$ (Paper 55) times the shared Hurwitz block in $\{\zeta(s, 1/4), \zeta(s, 3/4), \zeta(s, 5/4)\}$. Goncharov–Deligne 2005 faithfulness on $\pi_1^{\mathrm{mot}}(\mathbb{G}_m - \mu_4)$ would imply *full* $\mathbb{Q}$-linear independence of all N generators only for a rank-N per-sector map; the χ_{-4}-enriched content is rank 2 (parity-blind rank-1). Computing (rather than assuming) the $N(n_{\max}) \times N(n_{\max})$ Gram matrix of the parity-blind period-coefficient vectors gives **rank 1,** $\det = 0$ at every cutoff (`tests/test_paper56_injection_g4_periodmap.py`); the enriched χ_{-4} map is **rank** 2 (`tests/test_paper56_m3_parity_rank.py`; Remark 6.6), still $\ll N$, so the M3 column is not injective and lands in the abelianized image. (The earlier panel asserted a unimodular "standard basis" Gram with $\det = 1$; that was the hardcoded eye(N) the genuine computation refutes.) The depth-1 realisation of each generator in the Glanois basis $\mathcal{B}^4$ via the three quarter-integer Hurwitz blocks, with the Eskandari–Murty–Nemoto level-4 placement, remains valid as the description of the (collapsed) abelianized image.

Summing across the four panels gives the closed-form per-cutoff G4-Inj residual count

$$R^{\text{G4-Inj}}(n_{\max}) \;=\; (3N(n_{\max}))^2 + 3N(n_{\max}) + 5 + (1 + 3N(n_{\max})) \;=\; 9N(n_{\max})^2 + 6N(n_{\max}) + 6,$$

which evaluates to

$$R^{\text{G4-Inj}}(1) = 54, \quad R^{\text{G4-Inj}}(2) = 261, \quad R^{\text{G4-Inj}}(3) = 789, \quad R^{\text{G4-Inj}}(4) = 1854.$$

The per-cutoff breakdown is summarised in Table 2.

The extension from $n_{\max} = 2$ to $n_{\max} \in \{3, 4\}$ is structural rather than merely numerical: the closed-form per-cutoff identity makes the panel size depend on $N(n_{\max})$ only, with no $n_{\max}$-specific structural surprises. The four compatibilities remain bit-exact at every cutoff in the test grid, confirming that Theorem 6.2 is robust under refinement of the substrate cutoff. The pattern is $n_{\max}$-uniform, mirroring the cofiltered coherence of TC-2d (Theorem 5.13).

6.4 The natural realisation is Hodge / Hodge-de-Rham

The injection Φ^{inj} of Theorem 6.2 realises the *image* of the abelianization $(U^*_{GV})^{\mathrm{ab}}$ inside $\mathcal{U}_4^{\mathrm{ab}} \rtimes SL_2$ (it is *not* a closed immersion of U^*_{GV} itself, and the abelianization is not identified with a subgroup — the M3 column is rank 2, $\ll N$, so the map has a large kernel there). The motivic Galois group $\mathcal{G}_4$ admits multiple realisations — Betti, algebraic de Rham, Hodge / Hodge-de-Rham, étale, ℓ-adic, crystalline — and the comparison map at the period-ring level is realisation-dependent.

$n_{\max}$	$N(n_{\max})$	C1	C2	C3	C4	Total
1	2	36	6	5	7	54
2	5	225	15	5	16	261
3	9	729	27	5	28	789
4	14	1764	42	5	43	1854
	G4-Inj sum at $n_{\max} \in \{2,3,4\}$:					**2,904**

Table 2: Closed-form bit-exact residual counts for the level-4 cosmic-Galois injection panel at every finite cutoff $n_{\max} \in \{1,2,3,4\}$. The sum over the production grid $\{2,3,4\}$ contributes 2,904 residuals to the master verification panel (Table 3); the C4 row counts the *abelianized* M3 panel (its closed-immersion injectivity is refuted — Theorem 6.2). Every count is the explicit symbolic-zero output of `tests/test_paper56_injection_g4.py`; the closed-form $R^{\text{G4-Inj}}(n_{\max}) = 9N(n_{\max})^2 + 6N(n_{\max}) + 6$ matches each row bit-exact.

We adopt as the standing convention that the natural target for GeoVac's comparison program is the *Hodge / Hodge-de-Rham realisation*, not the étale or ℓ-adic realisations. Two structural reasons make this the only consistent choice:

1. **Substrate compatibility.** The real structure J with $J^2 = -\mathbf{1}$ (the Kramers / $SU(2) = S^3$ quaternionic structure) exists bit-exact at finite cutoff and supplies the complex structure a real-Hodge / Hodge-de-Rham realisation requires. We flag a finite-cutoff caveat (Sprint Hodge-SL_2, `debug/sprint_hodge_sl2_memo.md`): the clean odd KO-dimension-3 relation $JD = +DJ$ is *not* bit-exact on the natural substrate. With CG-weighted adjacency the off-diagonal part of D anticommutes with J exactly ($\{J, A_{CG}\} = 0$), but the m_j-independent diagonal Camporesi–Higuchi part *commutes* with J, so $D = \Lambda + \kappa A$ satisfies neither $JD = +DJ$ nor $JD = -DJ$ (the residual is exactly $2\Lambda J$). The complex structure is thus present — the structural input the Hodge realisation requires — while the clean odd KO-dimension-3 real spectral triple is only partially realised at finite cutoff. No comparable prerequisite for an étale or ℓ-adic realisation has been verified, nor is there a natural candidate at finite cutoff.

2. **Absence of p-adic input.** GeoVac has no p-adic place data, no Frobenius action, and no Galois group of a number field acting on the substrate. Étale / ℓ-adic realisations of motivic Galois groups require precisely this kind of arithmetic input; GeoVac supplies none of it. Categorically, those realisations are unreachable from GeoVac's data.

The level-4 injection $\Phi^{\text{inj}} : U^*_{GV} \to \mathcal{U}_4^{\text{ab}} \rtimes SL_2$ of Theorem 6.2 is therefore to be read as an *abelianized* Hodge / Hodge-de-Rham homomorphism (Reading A; not a closed immersion — the M3 column is rank 2, $\ll N$). The Hain–Brown relative completion of $SL_2(\mathbb{Z})$ carries a canonical mixed Hodge structure (Hain 2014 [6], Theorem 3.8); any future upgrade of the comparison target from $\mathcal{G}_4$ to a sub-quotient of the Hain–Brown relative completion would inherit that canonical MHS. At the time of writing, the Hain–Brown identification at the period level was probed at Sym^2 depth 1 and depth 2 and returned NEGATIVE (see `debug/sprint_hb_pslq_test_memo.md` and `debug/sprint_na1_depth2_mellin_memo.md`); the shape-match at the categorical level (extension $1 \to U \to G \to SL_2 \to 1$ with U pro-unipotent and tensor category generated by $\mathrm{Sym}^k V$) persists, but the comparison target remains $\mathcal{G}_4$ rather than Hain–Brown. The Hodge realisation choice is robust to either resolution.

From convention to identification: GeoVac realises a CM point. The Hodge realisation can be made concrete on the fundamental rep. Restricting the Kramers J to a $j = 1/2$ doublet of the lowest Dirac shell gives the canonical complex structure $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ on $V_{\text{fund}} = \mathbb{Q}^2$; with the SL_2-invariant symplectic form $Q = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ the Riemann form $QJ = I$ is positive-definite, so (V_{fund}, J, Q) is a genuine polarised weight-1 Hodge structure.

Proposition 6.7 (GeoVac realises the CM point; CM field $\mathbb{Q}(i)$)**.** *The Hodge group (the special Mumford–Tate group,* $\mathrm{MT} \cap SL_2$*) of GeoVac's canonical polarised weight-*1 *Hodge structure* (V_{fund}, J, Q) *is the norm-*1 *torus of* $\mathbb{Q}(i)$ *— a* 1*-dimensional CM torus inside* SL_2 *—* not *the full* SL_2*. (The full Mumford–Tate group adds only the weight scalars,* $\mathrm{Res}_{\mathbb{Q}(i)/\mathbb{Q}}\mathbb{G}_m$*; the CM conclusion is unchanged.)*

Proof. J is rational with $J^2 = -I$, so $\mathbb{Q}[J] \cong \mathbb{Q}[t]/(t^2+1) = \mathbb{Q}(i)$ and the Hodge circle $\cos t\, I + \sin t\, J$ generates the norm-1 torus $T_J = \{aI + bJ : a^2 + b^2 = 1\}$, with $\det(aI + bJ) = a^2 + b^2 = N_{\mathbb{Q}(i)/\mathbb{Q}}(a + bi)$. T_J is the $\mathbb{Q}$-Zariski closure of the Hodge circle (a 1-dimensional $\mathbb{Q}$-torus in which the real circle is dense) and preserves the definite Riemann form $QJ = I$, hence $T_J = SO(2)$ as a $\mathbb{Q}$-group, a proper 1-dimensional subgroup of SL_2 ($\dim T_J = 1 < 3 = \dim SL_2$). Equality of the Hodge group with SL_2 would require a non-algebraic (transcendental) complex structure, which the algebraic GeoVac substrate cannot produce. Verified bit-exact in `tests/test_paper56_hodge_sl2.py`. □

Remark 6.8 (Why this explains the Hain–Brown negative)**.** Proposition 6.7 upgrades the Hodge-realisation *convention* above to a concrete *identification*: GeoVac realises the complex-multiplication point of the weight-1 story, with CM field $\mathbb{Q}(i)$. This structurally explains the Hain–Brown period negative. Hain–Brown is the generic (full-SL_2, non-CM) modular locus, whose periods involve E_4, E_6, Δ; at a CM point the modular forms degenerate and the periods become CM-pure over the CM field. GeoVac's periods are therefore $\mathbb{Q}(i)$-pure (Catalan G, β-values), not modular — exactly the negative observed at Sym^2 depth 1 and depth 2.

Remark 6.9 (The $\mathbb{Q}(i)$ triangulation is spin-forced)**.** The CM field $\mathbb{Q}(i)$ coincides with the level-4 field $\mathbb{Q}(\mu_4)$ of $\mathcal{G}_4 = \mathcal{G}_{MT(\mathbb{Z}[i,1/2])}$ and with the field of GeoVac's period content (Catalan G mixed-Tate over $\mathbb{Q}(i)$, Eskandari–Murty–Nemoto 2025 [25]). A structural audit (`debug/sprint_hodge_sl2_aud` shows this is not a coincidence: both the Hodge CM field and the level-4 period content are gated by the half-integer spin structure. The complex structure exists iff $\varepsilon = (-1)^{2j} = -1$ (spinor; scalar $(-1)^{2l} = +1$ gives no complex structure), and the β/Catalan period content carries conductor $4 = |\mathrm{disc}\, \mathbb{Q}(i)|$ via the Dirac vertex-parity character χ_{-4} (Paper 28 [28]), whereas the scalar S^3 Laplacian ζ is pure-Tate over $\mathbb{Q}$ (conductor 1). The scalar null removes $\mathbb{Q}(i)$ from both sides at once. The motivic identification level-4 $= \mathbb{Q}(\mu_4)$ is taken from Deligne 2010 [7]; the audit verifies the common spin cause and the same-field identity, not the motivic result.

Remark 6.10 (Thermal time realises the Hodge circle; μ_4 at the quarter-periods)**.** The Hodge circle of Proposition 6.7 is not only abstractly present in the substrate — it is the compact thermal-time flow. On the $j = \frac{1}{2}$ doublet the modular generator of the truncated Bisognano–Wichmann wedge analysis is $K = \mathrm{diag}(2m_j)$, with integer spectrum, so $e^{2\pi i K} = \mathbf{1}$ exactly (the KMS $\beta = 2\pi$ circle). Transported to the Kramers frame U (the eigenframe of J), the flow *is* the Hodge circle, $U\, e^{itK}\, U^\dagger = \cos t\, \mathbf{1} + \sin t\, J$; in particular J is the flow's $\beta/4$ quarter-period point, the four quarter-periods are $\{\mathbf{1}, J, -\mathbf{1}, -J\} \cong \mu_4$ — the conductor $4 = |\mathrm{disc}\, \mathbb{Q}(i)|$ is the order of the flow's torsion point — and the half-period separates the sectors: $e^{i\pi K} = -\mathbf{1}$ on the spinor doublet (half-integer m_j; the spin double cover, visible inside the thermal circle) versus $+\mathbf{1}$ on any integer-l scalar sector, which factors through μ_2 and stays pure-Tate (Remark 6.9). All statements are exact (`tests/test_paper56_kms_hodge_circle.py`). Three qualifications keep this an Observation. First, any two maximal tori of $SU(2)$ are conjugate, and both halves of the statement are separately classical (the Hodge circle of a CM elliptic curve generates the norm-1 torus; the Bisognano–Wichmann flow is KMS at $\beta = 2\pi$); the content is the identification of the program's two *specific*, independently derived operators, and the location of the arithmetic data (μ_4 torsion, spinor sign) at canonical points of the flow. Second, compactness alone does not produce $\mathbb{Q}(i)$: the scalar sector also generates a compact circle yet its periods are pure-Tate over $\mathbb{Q}$. The discriminating datum is, however, *not* merely whether the μ_4 point acts faithfully (spinor) or through μ_2 (scalar): the Bargmann–Segal Hardy phase circle of the harmonic oscillator (Paper 24) carries a *faithful* μ_4—the metaplectic zero-point $\frac{3}{2}$ lifts the substrate's bipartition parity to order four—yet admits no CM Hodge structure, because its generator $N+\frac{3}{2}$ is strictly positive (no $\pm$-paired spectrum, hence no traceless 2-plane for a $\mathbb{Q}$-rational J to rotate). The discriminant separating the two carriers examined is a faithful μ_4 acting as a real rotation of a $\mathbb{Q}$-rational 2-plane, i.e. a $\pm$-symmetric generator spectrum—which the quaternionic spinor doublet supplies (Paper 24 reaches the same negative by three independent exact obstructions, of which this is one facet; no completeness or uniqueness claim is made). The conductor-4 *period* content, by contrast, does transfer: the Hardy tower's parity-graded spectral zeta is $2^{s-3}(\beta(s)-\beta(s-2))$, exactly one quarter of the Dirac value, with the χ_{-4} produced by the half-integer index shift times the parity grading (spin on S^3, metaplectic on S^5)—so an order-2 datum lifted to order 4 suffices for conductor-4 periods, while the CM *Hodge structure* additionally needs the $\pm$-paired rational carrier (`tests/test_paper24_hardy_circle_arithmetic.py`). Third, the 2π-closure re-reads an already-registered structural fact (the compactness of the truncated boost); it is a consistency statement, not independent evidence. We have not found this flow-level identification in the literature: the closest structures are the labelling of extremal KMS_∞ states by complex-multiplication points in the quantum-statistical-mechanics program of Connes–Marcolli–Ramachandran [30] (a labelling of states, not an identification of the flow) and a recent physical $U(1)\times U(1)$ R-symmetry flow realising a Hodge decomposition with complex multiplication at special loci [31] (an R-symmetry, not a thermal/modular flow).

Remark 6.11 (An independent chemistry realisation of the $\mathbb{Q}(i)$ arithmetic (Paper 59): convergent, not common-cause)**.** A concrete elliptic period from a wholly different corner of the program carries the same conductor-4 arithmetic — by a different forcing. The three-center electron-repulsion integral of Paper 59 [32] reduces, in momentum space, to a two-scale Bessel moment whose curve is the Legendre family over $X(2)=\Gamma(2)\backslash\mathfrak{H}^*$ (modulus $\lambda = 1-\rho$, with ρ the ratio of the two densities' Fock scales); its modular and $\mathbb{Q}(i)$-CM structure below is developed in the companion Paper 61 [33]. The load-bearing arithmetic statement is fibre-independent: the fibre period is $K(\lambda)=\frac{\pi}{2}\,\theta_3^2$, and θ_3^2 *is* the theta series of $\mathbb{Z}[i]$ — by Jacobi's two-squares theorem $r_2(n)=4\sum_{d|n}\chi_{-4}(d)$, the weight-one, level-4 Eisenstein series of character χ_{-4}, with $L(\theta_3^2,s)=4\,\zeta(s)\beta(s)$ — and the physical Bessel twist promotes the level from $\Gamma(2)$ to $\Gamma(4)$ (its decay rate makes $\sqrt{\rho}=\theta_4^2/\theta_3^2$ physical), whose field is $\mathbb{Q}(\mu_4)=\mathbb{Q}(i)$ (verified in `tests/test_paper59_theta_chi4.py`). The $\tau=i$ fibre at $\rho=\frac{1}{2}$, where the period is the $\mathbb{Q}(i)$-pure Γ-value $K(\frac{1}{2})=\Gamma(1/4)^2/(4\sqrt{\pi})$ (Chowla–Selberg, verified $\sim 10^{-51}$), remains a clean instance but is not load-bearing: any family sweeping its full real modulus locus passes through every CM fibre, the physically distinguished fibre is the coincident-scale cusp $\rho=1$ rather than $\rho=\frac{1}{2}$, and landing on the Legendre family is itself expected for a four-branch-point curve (Paper 59's own deflation). The group side sharpens this: the only involution the physics induces on the modulus is $\rho\mapsto 1/\rho$ (the density exchange; the observable descends to $X_0(2)$, Paper 59), whose physical fixed locus is the pure-Tate cusp $\rho=1$—so $\rho=\frac{1}{2}$ is one member of the exchange orbit $\{\frac{1}{2},2\}$, a visited fibre, not a fixed one. The two routes to $\mathbb{Q}(i)$ are therefore *convergent, not common-cause*: the Dirac route is spin-forced (half-integer shift $\times$ parity split $\to$ quarter-integer Hurwitz arguments $\to\chi_{-4}$, Remark 6.9) and mixed-Tate over $\mathbb{Q}(i)$, while the chemistry route is level-forced (the $\mathbb{Z}[i]$ theta series; $\Gamma(2)\to\Gamma(4)$) and mixed-elliptic with an irregular twist. They share the arithmetic pattern $4=2\times 2$ — a double-cover datum combined with a second $\mathbb{Z}_2$ — but the two double covers belong to different groups, and no comparison map between the two underlying $\mathbb{Q}^2$-carriers has been exhibited (the weight-$\frac{1}{2}$/spin analogy has a classical anchor: theta characteristics *are* spin structures, Atiyah 1971 [29]). The convergence is now quantified both ways: across the corpus's period-identification record, *blind* searches land on disc -4 in 0 of 30 independent campaigns (including two that targeted it), while *forced* constructions that exit the pure-Tate ring at all land on it in 3 of 3 — $\mathbb{Q}(i)$ is the first CM field the substrate's rational arithmetic can reach, a constrained-landscape statement rather than a coincidence (CHANGELOG v4.103.0); and on the modular side the arithmetic content of any single $\mathbb{Z}_2$ realisation is provably one bit ($\mathrm{PSL}_2(\mathbb{Z}) \cong \mathbb{Z}_2 * \mathbb{Z}_3$ torsion: order two forces disc -4, order three disc -3). Two concrete upgrades would convert convergence into mechanism: a comparison map *natural in the GeoVac data* — e.g. presenting the fundamental doublet as H_1 of a GeoVac-constructed abelian variety; note the polarisation-matching clause is void as a discriminator, since $h(\mathbb{Q}(i)) = 1$ makes the principally polarised rank-2 $\mathbb{Q}(i)$-CM Hodge structure unique up to isomorphism, so even a tautological identification carries $QJ = I$ to the Riemann form and only *functoriality* can discriminate — or a measured $\beta(2)$ in the physical integrated observable itself. The second test has now been *run*: an exact refactorization of the observable (Paper 59, the (k, w) frame) certified its value to 66 cross-validated digits, and the guarded, decoy-calibrated search is **decisively negative** for $\beta(2)$/Catalan content at height ≤ 10 through weight three (64 working digits) — no measured conductor-4 content exists in the integrated observable at clean heights, so, at reachable heights, the conductor-4 arithmetic remains a demonstrated property of the *family* (θ_3^2) and of the $D \to 0$ shadow, not of the physical value. (Calibration rider, correcting the earlier "decidable at ≥ 32 digits": height budgets scale as $10^{D/n}$ in the ring dimension n, so the dimension-twenty ring needed ~ 64 digits for even a height-10 verdict; the negative excludes clean closed forms, not large-height ones.) With T-1 tautology-blocked and T-2 negative at its reachable height, the seam stands as *convergent* on all tested legs. Whether the integrated observable, or its non-CM fibres, reaches the generic Hain–Brown locus remains open.

6.5 The equality direction is long-range

Theorem 6.2 closes the *injection direction* of the conjecture in Section 8.2. The complementary *equality direction* — whether Φ^{inj} is surjective onto $\mathcal{G}_4$ at the depth-graded level (or onto $\mathcal{U}_4^{\mathrm{ab}} \rtimes \mathbb{G}_m$ at the abelianization) — requires exhibiting (or ruling out) every level-4 cyclotomic motivic MZV as a GeoVac observable. At depth 1 all required Hurwitz values do appear in the M3 column; at depth ≥ 2 no GeoVac computation has yet produced an irreducible level-4 motivic MZV that is not reducible to depth-1 products. Whether this absence is structural (GeoVac's abelianized image is a strict, i.e. proper, sub-pro-algebraic subgroup of $\mathcal{G}_4$) or empirical (we have not looked hard enough) is the long-range forward research question — and it is testable in finite sprint-scale steps by exhaustion at increasing loop order in the master Mellin engine.

7 Verification panel

The full driver / data / memo trail of the verification panel is:

- PS-1: `debug/compute_q5p_ps1_transitions.py`, `debug/data/sprint_q5p_ps1_transitions.json`, `debug/sprint_q5p_ps1_transitions_memo.md`.
- PS-2: `debug/compute_q5p_ps2_ustar_compatibility.py`, `debug/data/sprint_q5p_ps2_ustar_compat` `debug/sprint_q5p_ps2_ustar_compatibility_memo.md`.
- PS-3: `debug/compute_q5p_ps3_inverse_limit.py`, `debug/data/sprint_q5p_ps3_inverse_limit.json` `debug/sprint_q5p_ps3_inverse_limit_memo.md`.
- PS-4: `debug/compute_q5p_ps4_endo_rigidity.py`, `debug/data/sprint_q5p_ps4_endo_rigidity.json` `debug/sprint_q5p_ps4_endo_rigidity_memo.md`.
- TC-1a/b/c/d/e/f: `debug/compute_q5p_tc1{a, b, c, d, e, f}_*.py`, `debug/data/sprint_q5p_tc1{a` `b, c, d, e}_*.json` + `debug/data/q5p_tc1f_sl2_inclusion.json`, + memos in `debug/sprint_q5p_tc` `b, c, d, e, f}_*.md`.

Sub-track	Statement	Residuals
PS-1	Transitions $\Phi_{m,k}$ + cofiltered axiom (PS-1)	130
PS-2	U^*-equivariance ($\mathbb{G}_a$ + class-level) (PS-2)	485
PS-3	Inverse limit + continuity (PS-3)	284
PS-4	Endomorphism rigidity (PS-4)	872
TC-1a	$\mathrm{Rep}_{\mathrm{fin}}$ abelian	106
TC-1b	$\mathrm{Rep}_{\mathrm{fin}}$ symmetric monoidal	56
TC-1c	$\mathrm{Rep}_{\mathrm{fin}}$ rigid	50
TC-1d	Fiber functor ω exact + faithful + $\otimes$ + unit	40
TC-1e	$\mathbb{G}_a$ inclusion (natural $\otimes$-auto axioms + group law)	98
TC-1f	SL_2 axioms on $\mathfrak{PW}$ panel	135
TC-1f	$\mathbb{G}_a \times \mathrm{SL}_2$ commute on combined reps	100
TC-1f	Full-panel injectivity at $n_{\max} = 2$	255
TC-2a	Finite-cutoff equality on abelian factor at $n_{\max} = 2$	16
TC-2b	SL_2-factor equality (Sym^2 + det + decomp + dim)	17
TC-2c	Per-cutoff equality at $n_{\max} = 3, 4$ (Φ-recovery + dim + η_T)	73
TC-2d	Cofiltered coherence (panel + transitivity + group-hom)	243
G4-Inj	Level-4 injection (C1 + C2 + C3 + C4 at $n_{\max} = 2$)	261
G4-Inj	Level-4 injection at $n_{\max} = 3$ (C1 + C2 + C3 + C4)	789
G4-Inj	Level-4 injection at $n_{\max} = 4$ (C1 + C2 + C3 + C4)	1854
Total bit-exact zero residuals		**5,864**

Table 3: The full Tannakian-substrate verification panel: 5,864 bit-exact zero residuals across the four pro-system sub-tracks (PS-1/2/3/4), the six Tannakian-foundation sub-tracks (TC-1a/b/c/d/e/f), the four converse-direction sub-tracks (TC-2a/b/c/d), and the level-4 cosmic-Galois injection panel extended to every finite cutoff in the production grid $n_{\max} \in \{2, 3, 4\}$ (G4-Inj $\times 3$). All residuals computed in `sympy`-rational arithmetic; no floating-point steps anywhere in the verification chain.

- TC-2a: `debug/compute_q5p_tc2a_aut_equality.py`, `debug/data/sprint_q5p_tc2a_aut_equality.jso` `debug/sprint_q5p_tc2a_aut_equality_memo.md`.
- TC-2b: `debug/compute_q5p_tc2b_sl2_aut_equality.py`, `debug/data/sprint_q5p_tc2b_sl2_aut_equa` `debug/sprint_q5p_tc2b_sl2_aut_equality_memo.md`.
- TC-2c: `debug/compute_q5p_tc2c_higher_cutoff.py`, `debug/data/sprint_q5p_tc2c_higher_cutoff.j` `debug/sprint_q5p_tc2c_higher_cutoff_memo.md`.
- TC-2d: `debug/compute_q5p_tc2d_cofiltered_coherence.py`, `debug/data/sprint_q5p_tc2d_cofilter` `debug/sprint_q5p_tc2d_cofiltered_coherence_memo.md`.

The module `geovac/tannakian.py` (TC-1a..TC-1f, ∼2,400 lines) and `geovac/pro_system.py` (PS-1..PS-4) implement the substrate; both are bit-exact `sympy.Rational` / `Integer` throughout. Test coverage: 282 tests pass (49 base Tannakian + 30 rigidity + 24 fiber functor + 25 TC-1e + 60 TC-1f + 61 pro-system + 9 TC-2a + 14 TC-2b + 3 TC-2c fast + 7 TC-2d; plus 1 TC-2c slow, $n_{\max} = 4$, run separately).

8 Open questions and frontier

8.1 The converse direction — closed at finite cutoff (TC-2)

The converse direction of Deligne–Milne 1982 Theorem 2.11 on the finite-cutoff substrate, $\mathrm{Aut}^{\otimes}(\omega) = U^*_{\mathrm{Levi}}$ at each $n_{\max}$, is closed by the TC-2 sub-arc (Theorems 5.7, 5.9, 5.11, 5.13). The substantive open content is the *inverse-limit identification at the natural cosmic-Galois comparison target.*

8.2 The natural cosmic-Galois comparison target

The comparison target for $\mathrm{Aut}^{\otimes}(\omega)^{(\infty)}$ is the level-4 motivic Galois group

$$\mathcal{G}_4 = \mathcal{G}_{MT(\mathbb{Z}[i,1/2])} = \mathbb{G}_m \ltimes \mathcal{U}_4,$$

of Deligne 2010 [7] and Glanois 2015 [8], *not* Brown's smaller $\mathcal{G}_{MT(\mathbb{Z})}$. Three published results motivate this refinement: Paper 55 [23] classifies GeoVac's period ring inside $\mathrm{MT}(\mathbb{Z}[i, 1/2])$ at level ≤ 4; Eskandari–Murty–Nemoto 2025 [25] construct a two-dimensional motive carrying Catalan $G = \beta(2)$, prove that *this motive* is not mixed Tate over $\mathbb{Q}$, and show it becomes mixed Tate after base change to $\mathbb{Q}(i)$—which is where G is known to live, and matters here because GeoVac's M3 vertex-parity sector produces G in QED vertex sums. (The stronger statement, that G is not a period of *any* mixed Tate motive over $\mathbb{Q}$, is expected but open, so this motivates rather than forces the level.) And the Goncharov–Deligne 2005 faithfulness of $\mathcal{G}_N$-action on $\pi_1^{\mathrm{mot}}(\mathbb{G}_m - \mu_N)$ supplies the faithful action underlying the level-4 comparison. (Injectivity of that comparison would require a rank-N M3 period map, which Theorem 6.2 refutes; the χ_{-4} vertex parity caps the rank at 2.)

> **Theorem (cosmic-Galois injection, this paper).** The injection direction is an *abelianized homomorphism* $U^*_{GV} \to \mathcal{U}_4 \rtimes SL_2$ — *not* an injective closed immersion (the M3 column is rank 2, capped, $\ll N$) — realising the abelianized image as a closed subgroup at the abelianization-of-unipotent level; this is now *closed at theorem grade*: see Theorem 6.2 in Section 6 for the explicit period map, the four structural compatibilities (multiplicativity, depth-1 coproduct compatibility, SL_2-to-Levi via χ_4, abelianized image via Glanois basis), with the M3 column at rank 2 ($\ll N$) so the map is the abelianized homomorphism rather than an injective closed immersion (Goncharov–Deligne 2005 faithfulness on $\pi_1^{\mathrm{mot}}(\mathbb{G}_m - \mu_4)$ would apply only to a sector-resolved rank-≥ 2 M3 period map; the χ_{-4} vertex parity now supplies one at rank 2, capped there, so it refines but does not reach a closed immersion — Remark 6.6).
>
> **Remaining conjecture (equality / exhaustion).** Equality $\Phi^{\mathrm{inj}}(U^*_{GV}) = \mathcal{U}_4^{\mathrm{ab}} \rtimes SL_2$ holds iff every level-4 cyclotomic motivic MZV at depth ≥ 2 is realised by some GeoVac observable.

The injection direction is now a theorem (Section 6). The equality / exhaustion direction is the long-range forward research question: no depth-≥ 2 level-4 cyclotomic MZV has yet been observed in any GeoVac computation. Whether this is because we have not looked, or because GeoVac is structurally a strict sub-pro-algebraic group of $\mathcal{U}_4^{\mathrm{ab}}$, is the named open content past injection (Remark 6.5 delimits scope).

A common framing slip worth flagging: the two groups are related by a *surjection* $U^*_{CM} \twoheadrightarrow \mathcal{G}_{MT(\mathbb{Z})}$, not an identification — an elementary comparison of two free pro-unipotent groups rather than an imported theorem, and one Brown himself declines to assert. Connes–Marcolli's U^*_{CM} has one generator in every positive integer degree (free non-abelian) while $\mathcal{G}_{MT(\mathbb{Z})}$ has generators only in odd degrees ≥ 3; the surplus is the cancellation of even-zeta values $\zeta(2k) \in \pi^{2k}\mathbb{Q}$ already accounted for by the $\mathbb{G}_m$ factor. The GeoVac–cosmic-Galois comparison should be stated against $\mathcal{G}_4$, with the relationship to U^*_{CM} obtained via the (level-1) sub-quotient.

8.3 Abelian vs free non-abelian unipotent (NA-1 structural finding)

A structural feature distinguishes U^*_{GV} from the motivic-Galois target $\mathcal{U}_4$: GeoVac's unipotent radical is *abelian* ($\mathbb{G}_a^\infty$) while the motivic unipotent radical is *free non-abelian*. This is not an accidental commutation of generators; by the Cartier–Milnor–Moore theorem in characteristic 0, any primitive Hopf algebra $\mathrm{Sym}_\mathbb{Q}(V)$ has abelian Tannakian dual by construction. For non-abelian content the substrate would need to be enriched to the cofree-cocommutative shuffle Hopf algebra $T(V)$ with deconcatenation coproduct — the motivic-style structure (Brown 2012 [3]). Two structural readings remain consistent with current evidence:

- **Reading A (abelianization).** GeoVac respects the primitive coproduct, and U^*_{GV} is the *abelianization* of the motivic free non-abelian unipotent — a specific weight-only geometric realisation that drops the depth/iteration content.
- **Reading B (shuffle enrichment).** GeoVac's physical content (Mellin moments, master Mellin engine products) naturally respects the shuffle coproduct, and the substrate should be enriched to $T(V)$, at which point U^*_{GV} upgrades to a free non-abelian unipotent.

A sprint-scale concrete test distinguishes the readings: compute the joint Mellin transform of two specific depth-1 moments (M2 Seeley–DeWitt × M3 vertex-parity Hurwitz) and check whether the result factorises symmetrically (Reading A) or asymmetrically / deconcatenation-pair (Reading B). Sprint memo `debug/sprint_q5p_na1_non_abelian_probe_memo.md`.

Empirical close — Reading A wins (Sprint JLO-Depth2, 2026-06-07). The naive Mellin-trace test above turns out to be *structurally insensitive* to the A/B distinction. Sprint NA-1 (2026-06-06) demonstrated that the depth-2 joint Mellin transform $J(s_1, s_2)$ collapses bit-exactly to a depth-1 parity-graded Mellin moment at shifted argument, $J(s_1, s_2) = M_3^{\gamma_P}(s_1+s_2-1)$, on the natural Camporesi–Higuchi diagonal substrate (Paper 55 [23] Theorem `thm:na1_diagonal_collapse`). Sprint NA-1-offdiag (2026-06-06) generalised this: the trace-functional collapse holds on *any* substrate, commuting or non-commuting, because the trace functional is blind to off-diagonal entries of inserted operators in the D-eigenbasis (Paper 55 [23] Theorem `thm:na1_trace_functional_collapse`, "Reading C-strong"). Substrate enrichment cannot recover the A/B distinction at the trace-of-single-γ-insertion level.

The right next probe is the JLO entire cyclic cocycle at depth 2, $\chi_D(a_0, a_1, a_2)$, where two non-D-function insertions between three heat-kernel factors preserve the off-diagonal information the trace functional discards. Sprint JLO-Depth2 (`debug/sprint_jlo_depth2_memo.md`, 2026-06-07) returns **Reading A bit-exact**: on the truncated Camporesi–Higuchi spectral triple at $n_{\max} \in \{2, 3\}$ with the sector-idempotent algebra $\mathcal{A} = \mathbb{C}\langle e_0, \ldots, e_{N-1}\rangle$, the cocycle $\chi_D(a_0, a_1, a_2; t)$ is bit-exactly S_3-invariant under permutation of (a_0, a_1, a_2) at both leading order (t^0) and next order (t^1), in both the even (chirality-graded) and odd (plain trace) flavours. The 25-pair $\{1, e_s, e_t\}$ panel returns 40 bit-exact zero swap-asymmetric residuals at $n_{\max} = 2$; the full S_3-orbit test on $\chi_D(e_0, e_1, e_2)$ returns the same value across all six permutations. See Paper 55 [23] Theorem `thm:jlo_depth2_reading_A`.

The depth-2 content sits entirely in the symmetric / primitive-product sector of $S_3 \curvearrowright \mathcal{A}^{\otimes 3}$, with zero antisymmetric / shuffle-Hopf content. The honest-scope caveat: while commutativity of $\mathcal{A}$ plus trace cyclicity force *some* permutation invariance, the operator-level commutators that would reduce S_3-invariance to mere cyclicity are non-zero on this substrate (e.g. $\|[\gamma, D]\| = \sqrt{13}/4$, $\|[[D, e_0], [D, e_1]]\| = 3/64$ at $n_{\max} = 2$), so the bit-exact S_3-cocommutativity is strictly stronger than what those weaker arguments give.

Strategic implication. On the natural substrate, GeoVac's JLO-cocycle-level content maps onto the abelianisation-by-Cartier–Milnor–Moore of the cosmic-Galois universe. The map of Theorem 6.2, $U^*_{\mathrm{GV}} \to \mathcal{U}_4^{\mathrm{ab}} \rtimes SL_2$, is the *structurally correct* statement of what is available—not a partial version of a closed immersion. It is emphatically *not* injective on the sector-indexed generators (the M3 column is rank 2, $\ll N$), so what is realised is the abelianized *image*, not an embedding of U^*_{GV}. Reading B (shuffle Hopf $T(V)$ enrichment) is *ruled out empirically at depth 2* on the natural substrate; no sprint-scale follow-on to a substantial free non-abelian Tannakian closure is empirically demanded. Two open questions remain: (i) non-commutative inner-factor enrichment of $\mathcal{A}$ could expose Reading B content at higher Mellin slot indices, but Sprint H1's Yukawa-non-selection theorem (§6.4 above and Paper 32 [16] §VIII.C) already classifies the inner factor as calibration data, not autonomously generated by the GeoVac outer triple, so this is not a sprint-scale follow-on; (ii) the equality direction of Theorem 6.2 (whether Φ^{inj} is surjective onto $\mathcal{U}_4^{\mathrm{ab}} \rtimes SL_2$) remains the named long-range forward research question (§6.5), testable by exhausting GeoVac's M3 sector for level-4 cyclotomic motivic MZVs at depth ≥ 2.

8.4 Pro-finite assembly (long-range)

The full long-range content of the cosmic-Galois closure is the pro-finite Tannakian theorem coherent with v3.66.0 FO3 Interpretation C at the period level on $\mathcal{O}_\infty$ (PS-3, Theorem 2.5). This identifies the inverse-limit $\mathrm{Aut}^{\otimes}(\omega_{\mathcal{O}_\infty}) = \mathbb{G}_a^\infty \rtimes SL_2$ with (a sub-quotient of) $\mathcal{G}_4$ at the period level, in the spirit of Brown 2012 [3], Deligne 2010 [7], and Glanois 2015 [8].

8.5 Sprint-scale follow-ons

- **Higher-$n_{\max}$ injectivity panels.** Theorem 5.4 at $n_{\max} \in \{3, 4, 5\}$: same construction, panel size grows linearly in $n_{\max}$ as $3 \cdot N(n_{\max})$. Sprint-scale.
- SL_2 **on higher Peter–Weyl reps.** Theorem 5.2 on $\mathrm{Sym}^k V_{\mathrm{fund}}$ for $k \geq 3$. Sprint-scale.
- **Non-natural-substrate enlargements.** The Peter–Weyl decoration of §5.2 is the natural substrate (acts on disjoint tensor factor from $\mathbb{G}_a^{3N}$, so commutativity is by the Kronecker identity). Genuinely-new strong-form content lives on *enlarged* substrates where $\mathbb{G}_a$ and SL_2 act on the same tensor factor; that is long-range structural content, analogous to the Paper 46 Appendix B enlarged-substrate program.
- **Inverse-limit reconstruction.** The reconstruction argument at the inverse-limit level $\mathcal{O}_\infty$ coherent with the v3.66.0 FO3 period content is named as PS-3.5 in the roadmap; long-range, the heart of Tannakian reconstruction for GeoVac.

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References

[1] P. Deligne and J. S. Milne, *Tannakian categories*, in "Hodge Cycles, Motives, and Shimura Varieties", Lecture Notes in Math. **900** (1982) 101–228.

[2] P. Deligne, *Catégories tannakiennes*, in "The Grothendieck Festschrift" Vol. II, Progr. Math. **87** (Birkhäuser, 1990) 111–195.

[3] F. Brown, *Mixed Tate motives over $\mathbb{Z}$*, Ann. of Math. **175** (2012) 949–976.

[4] Y. André, *Une introduction aux motifs (motifs purs, motifs mixtes, périodes)*, Panoramas et Synthèses **17**, Société Mathématique de France, 2004.

[5] F. Brown, *Motivic periods and the projective line minus three points*, Proc. ICM 2014, Vol. II (Kyung Moon Sa, Seoul, 2014) 295–318; arXiv:1407.5165.

[6] R. Hain, *The Hodge-de Rham theory of modular groups*, in "Recent advances in Hodge theory" (LMS Lecture Notes 427, Cambridge Univ. Press, 2016) 422–514; arXiv:1403.6443.

[7] P. Deligne, *Le groupe fondamental unipotent motivique de* $\mathbb{G}_m - \mu_N$*, pour* $N = 2, 3, 4, 6, 8$, Publ. Math. IHÉS **112** (2010) 101–141.

[8] C. Glanois, *Periods of the motivic fundamental group of* $\mathbb{P}^1 - \{0, \mu_N, \infty\}$, PhD thesis, Université Pierre et Marie Curie, 2015.

[9] A. Connes and M. Marcolli, *Renormalization and motivic Galois theory*, Internat. Math. Res. Notices **2004(76)** (2004) 4073–4091; arXiv:math/0409306.

[10] M. Marcolli and W. D. van Suijlekom, *Gauge networks in noncommutative geometry*, J. Geom. Phys. **75** (2014) 71–91; arXiv:1301.3480.

[11] C. I. Perez-Sanchez, *Bratteli networks and the Spectral Action on quivers*, arXiv:2401.03705 (2024).

[12] C. I. Perez-Sanchez, *Comment on 'Gauge networks in noncommutative geometry'*, arXiv:2508.17338 (2025).

[13] J. Loutey, *The Dimensionless Vacuum: Recovering the Schrödinger Equation from Scale-Invariant Graph Topology*, GeoVac Paper 7 (2025).

[14] J. Loutey, *Spectral-Geometric Exchange Constants: Projecting Discrete Spectra onto Continuous Manifolds*, GeoVac Paper 18 (2025–2026).

[15] J. Loutey, *The Universal/Coulomb Partition of the GeoVac Spectral Triple*, GeoVac Paper 31 (2026).

[16] J. Loutey, *The GeoVac Spectral Triple: A Synthesis Paper Locking the Framework into the Marcolli–van Suijlekom Gauge-Network Lineage*, GeoVac Paper 32 (2026).

[17] J. Loutey, *State-space Gromov–Hausdorff convergence of truncated Camporesi–Higuchi spectral triples on* S^3, GeoVac Paper 38 (2026).

[18] J. Loutey, *Lorentzian propinquity on truncated* $SU(2) \times U(1)_T$ *Krein spectral triples: a degeneracy theorem*, GeoVac Paper 45 (2026).

[19] J. Loutey, *Strong-form Lorentzian propinquity on truncated* $SU(2) \times U(1)_T$ *Krein spectral triples*, GeoVac Paper 46 (2026).

[20] J. Loutey, *Norm-resolvent convergence to the non-compact Lorentzian Krein spectral triple on* $S^3 \times \mathbb{R}_t$, GeoVac Paper 47 (2026).

[21] J. Loutey, *The Krein–Mondino–Sämann bridge:* K^+*-weak-form synthesis*, GeoVac Paper 48 (2026).

[22] J. Loutey, *Operator-system Lorentzian pre-length spaces and the strong-form Krein–Mondino–Sämann bridge*, GeoVac Paper 49 (2026).

[23] J. Loutey, *Periods of GeoVac: a cyclotomic mixed-Tate classification of the master Mellin engine*, GeoVac Paper 55 (2026).

[24] F. Fathizadeh and M. Marcolli, *Periods and motives in the spectral action of Robertson–Walker spacetimes*, Comm. Math. Phys. **356** (2017) 641–671; arXiv:1611.01815.

[25] P. Eskandari, V. K. Murty, and Y. Nemoto, *Mixed motives and linear forms in the Catalan constant*, arXiv:2510.20648 (2025).

[26] A. B. Goncharov, *Periods and mixed motives*, preprint arXiv:math/0202154 (2002); see also P. Deligne and A. B. Goncharov, *Groupes fondamentaux motiviques de Tate mixte*, *Ann. Sci. École Norm. Sup.* (4) **38**, 1–56 (2005). The Goncharov–Deligne 2005 result establishes the faithful action of the motivic Galois group $\mathcal{G}_{\mathrm{MT}(\mathcal{O}_N[1/N])}$ on the motivic fundamental group $\pi_1^{\mathrm{mot}}(\mathbb{G}_m - \mu_N)$, used here as the injection-direction backbone for the level-4 period comparison.

[27] J. W. Milnor and J. C. Moore, *On the structure of Hopf algebras*, *Ann. of Math.* (2) **81**, 211–264 (1965); P. Cartier, *A primer of Hopf algebras*, in *Frontiers in Number Theory, Physics, and Geometry II*, Springer (2007) 537–615. The Cartier–Milnor–Moore theorem in characteristic zero identifies any primitively generated cocommutative Hopf algebra as the universal enveloping algebra of its Lie algebra of primitives; applied to $\mathcal{H}_{\mathrm{GV}} = \mathrm{Sym}_{\mathbb{Q}}(V)$ (with V abelian), the Tannakian dual is forced to be abelian unipotent (NA-1 structural finding).

[28] J. Loutey, *QED on S^3: Spectral Geometry of Perturbative Transcendentals*, GeoVac Paper 28 (2026).

[29] M. F. Atiyah, *Riemann surfaces and spin structures*, Ann. Sci. Éc. Norm. Supér. (4) **4** (1971) 47–62.

[30] A. Connes, M. Marcolli, and N. Ramachandran, *KMS states and complex multiplication*, Selecta Math. (N.S.) **11** (2005) 325–347; arXiv:math/0501424.

[31] R. Angius and R. Volpato, *Hodge loci and complex multiplication via generalized symmetries in Calabi–Yau sigma models*, arXiv:2605.30418.

[32] J. Loutey, *The Three-Center Electron-Repulsion Integral Is an Elliptic Bessel Moment: Genus One at the Third Center in Momentum-Space Slater Theory*, GeoVac Paper 59 (2026).

[33] J. Loutey, *The Bessel-Moment Period Algebra of the Three-Center Coulomb Integral: Modular Structure and Cosmic-Galois Placement*, GeoVac Paper 61 (2026).